Mixed-model association under population structure: numerical foundations, calibration diagnostics and research directions for mixmogam

mixmogam contributors

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Historical studies retain their original versions; the new benchmarks use 2.0.0.dev8

Abstract

We ask how population structure exposes errors in mixed-model association tests, using the mixmogam package: dense and streaming leave-one-chromosome-out (LOCO) scans and HRATT, a two-step test inspired by LDAK-KVIK. The association denominator depends on a variant's orientation in the kinship spectrum, so a scalar calibration that is right on average can be wrong in opposite directions for different variants. In a known-covariance experiment with 256 samples and 970 held-out audit variants, scalar calibration under moderate frequency dispersion rejected 1.00% at a nominal 1%, but 1.19% and 0.73% in the lowest and highest structure-loading quintiles; a rank-four spectral correction gave 1.010% and 1.005%. In matched simulations at $n = 800$ with LD, an omitted environmental effect of ancestry gave official LDAK-KVIK 15.83% rejection under the genetic null at 1%, and HRATT 0.64%; two genotype PCs brought these to 0.97% and 0.92%. We extended HRATT to case-control outcomes and sampling weights. In prespecified simulations the weighted test stayed calibrated under selection on the outcome, where the sandwich variance rejected up to 26 times too often; cross-fitted LOCO scores brought effect bias within 2% in most cells; and covariate-specific genotype distributions with principal components removed inflation from ancestry-dependent weights. Two criteria failed: weighted effects under selection on the outcome were 5.4% low (Monte Carlo standard error 2.1%), and weighted fits took 1.55 times as long as HRATT-HE against a target of 1.3. On the *Arabidopsis thaliana* RegMap panel every method found six of seven known major genes; none found MOT1 on the prespecified raw scale, so the check failed (an exploratory log-scale rerun ranked MOT1 first). At 100,000 samples and 20,000 markers, 2.0.0.dev8 took a median 25.84 seconds for HRATT-HE, 37.59 for LDAK-KVIK and 101.77 for BOLT-inf. Apart from the RegMap check all evidence is simulated, the new timings come from one workstation, and some variational fits did not converge under strong confounding.

1 Introduction

A mixed-model association test asks whether a candidate variant explains phenotypic variation beyond specified covariates and a model of genetic background. Its appeal is that the same relationship matrix can describe close relatives, cryptic relatedness, and broad population structure. Early efficient implementations made this approach practical in model organisms and human association studies (Kang et al., 2008, 2010; Lippert et al., 2011; Zhou and Stephens, 2012). The resulting test is nevertheless conditional on assumptions about the phenotype mean, the background covariance, and the way nuisance parameters are fitted. A precise matrix calculation cannot repair a misspecified mean, and a good prediction model is not by itself a valid association test.

The computational problem has changed with cohort size. Dense methods factorize an $n \times n$ covariance matrix and reuse the factorization across variants. Large-cohort methods instead use iterative fitting, whole-genome prediction, and approximate association statistics. BOLT-LMM combines mixed-model association with a mixture prior on marker effects (Loh et al., 2015); REGENIE constructs whole-genome predictions through successive ridge regressions (Mbatchou et al., 2021); and newer approaches include LDAK-KVIK (Hof and Speed, 2025), which inspired this package's Heritability-weighted Residual Association Two-step Test (HRATT), and the variational method Quickdraws (Loya et al., 2025). Their fitting and testing procedures differ, so a finding about one denominator approximation applies to that approximation, not to every two-step method.

Two statistical choices cut across these computational designs. First, a candidate variant and its correlated neighbours can contribute to the random-effect predictor against which they are tested. This proximal contamination can reduce power; excluding the candidate chromosome or region changes that overlap (Listgarten et al., 2012; Yang et al., 2014). Second, ancestry can influence both allele frequencies and environmental exposures. Modelling genetic covariance and adjusting the phenotype mean with principal components (PCs) address different parts of this problem (Price et al., 2006; Vilhjálmsson and Nordborg, 2013). Neither chromosome exclusion nor covariance modelling removes environmental confounding.

This report is about a third, narrower issue. The denominator of an association statistic depends on how the tested variant aligns with the covariance eigenvectors. Replacing it by a scalar multiple of the variant norm works when that alignment matters little. Under structure, a good average can hide inflation in one class of variants and deflation in another. What matters is the conditional rejection probability for a variant, or a prespecified class of variants, under a stated null model, and a genome-wide median does not measure it.

mixmogam is a convenient place to study these distinctions because its dense and streaming calculations share genotype handling and several model components. It began with multi-locus mixed-model association (Segura et al., 2012) and now also provides LOCO and two-step analyses. We set out the mathematical contracts needed to interpret its outputs, run a known-covariance experiment with a disjoint set of audit variants, compare complete scans with the official LDAK-KVIK executable on matched phensim simulations with and without population structure and LD, and reread archived simulations. Section 2.12 extends HRATT to case-control outcomes and sampling weights. We close with research questions that could tell a general method from an approximation that works only in favourable examples.

The evidence comes at several levels: software checks of numerical and data-handling contracts, the known-covariance experiment, the matched benchmark, larger fixed-panel experiments that separate estimator choice, parallelism and storage, and historical simulations that predate the correctness review. Table 1 lists what each supports.

2 Statistical and computational methods

2.1 Data, estimand, and relationship matrix

Let $\mathbf{y} \in \mathbb{R}^n$ be a quantitative phenotype (binary outcomes and sampling weights enter HRATT through Section 2.12), $\mathbf{X}$ a full-rank $n \times p$ covariate matrix including an intercept, and $\mathbf{z}_j$ a centered candidate genotype. The coefficient of interest, α_j, is the fixed effect in a specified marginal mixed model after integrating out the random genetic background. It is not automatically a direct causal effect: linkage disequilibrium (LD), omitted exposures, and the choice of background model affect its interpretation. The null is $H_{0j} : \alpha_j = 0$ under that model. A variant without a designated large QTL effect need not satisfy this null if a simulated polygenic background still assigns it an effect.

Table 1: Evidence used in this revision and the inference it supports. Historical archives keep their original versions.

Evidence	Established here	Remaining boundary
Initial review, 2.0.0.dev1	Regression and independent numerical checks; 117 local tests passed; installed-artifact checks	Test coverage does not establish population calibration.
New denominator experiment	Analytic conditional rejection probabilities, checked by independent phenotype replicates	Known Gaussian covariance, small fixed genotype panels, exact leading eigenvectors.
Archived simulations	Descriptive patterns under the recorded historical implementation	New input handling and fitting changes need fresh end-to-end runs.
Matched phensim benchmark	Independent BED decoding and reference-reader checks; null calibration, QTL detection and process resource measurements; mixmogam methods rerun with 2.0.0.dev6 on the same inputs	Six independent panels per cell; small quantitative-trait simulations, controlled LD and three discrete populations; official timings from three days earlier.
Computational extension, 2.0.0.dev3 and later reruns	Projected-moment validation; paired implementation, thread and storage comparisons; at 50,000 samples and 20,000 markers, 12 cross-fitted mixmogam fits with threaded fold products, 12 with in-sample scores, 12 storage fits and 12 official fits	Two fixed timing panels; different variance estimators across programs; strict thread-tolerance failures retained; official timings from a day when identical mixmogam code took 38% longer.
Work reduction, 2.0.0.dev5	Paired fresh-process fits against frozen 2.0.0.dev4 on five workloads, with result agreement; a same-day check at 50,000 samples; Lanczos steps and probes against dense REML on six panels	Two repetitions on a loaded host; simulated panels; two-step results move by Monte Carlo amounts.
Genotype streaming, commits 12aa504 to e4089d8	Paired fits of 2.0.0.dev5 against commit d51b4c3 on every path, three repetitions gated on host load, with result agreement; archives of each step; a same-day check at 50,000 samples	Simulated panels; one host.
Case-control and weighted HRATT	Prespecified simulation: 30 null and 10 mixed replicates per cell, six selection scenarios, prevalence 1–20%, with official LDAK-KVIK	Independent markers and discrete populations; two of seven criteria passed.
Cross-fitted scores and covariate-specific genotype distributions	Prespecified simulation on the same samples, with $F_{ST} = 0.05$ cells with and without principal components; 660 cases; the principal-component cells rerun with the built-in components (130 cases)	Five of seven criteria passed; weighted effects 5.4% low under selection on the outcome.
Archived external comparison	A provenance failure was identified: incorrect PLINK export encoding	External-program calibration, agreement, and superiority claims from that archive remain withdrawn.

The relationship matrix summarizes the covariance induced by a polygenic marker model (Yang et al., 2010). For the unweighted matrix, mixmogam centers each marker by its mean among called samples and divides by its empirical standard deviation among those samples. Missing calls become zero after this transformation, corresponding to mean imputation. Write the resulting background matrix as $\mathbf{Z}$, with M retained polymorphic markers. Its raw relationship matrix and a weighted extension are

$$\mathbf{K}_0 = \frac{1}{M}\mathbf{Z}\mathbf{Z}^\top, \qquad \mathbf{K}_w = \frac{\sum_{j=1}^{M} w_j \mathbf{z}_j \mathbf{z}_j^\top}{\sum_{j=1}^{M} w_j}, \quad w_j \geq 0, \quad \sum_j w_j > 0. \tag{1}$$

Empirical-standard-deviation scaling differs from using the Hardy–Weinberg variance $2f_j(1-f_j)$; these definitions should not be interchanged in a reference comparison. The optional kinship rescaling subtracts the mean off-diagonal entry from every entry and divides by the resulting mean diagonal. An intercept absorbs the constant-matrix shift in the residual subspace. The raw, unrescaled matrix is used in the new experiment in Section 3.1.

The current input contract is diploid hard calls $0, 1, 2$, with -1 for missing values. Minor allele count is measured in allele copies. Sample identifiers must align uniquely; silently intersecting phenotype and genotype rows in different orders changes the scientific problem. PLINK effect-allele orientation, missingness, and identifier preservation are therefore part of the statistical contract.

2.2 Gaussian model, variance fitting, and chromosome exclusion

For a candidate in chromosome group g, the working model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{z}_j\alpha_j + \mathbf{u} + \mathbf{e}, \quad \mathbf{u} \sim N(0, \sigma_g^2 \mathbf{K}_{-g}), \quad \mathbf{e} \sim N(0, \sigma_e^2 \mathbf{I}), \quad \mathbf{u} \perp \mathbf{e}. \tag{2}$$

Here $\mathbf{K}_{-g}$ is constructed without the markers in group g. The subscript on σ_g^2 denotes genetic variance, not the group index. With $\delta = \sigma_e^2/\sigma_g^2$, set $\mathbf{H} = \mathbf{K}_{-g} + \delta\mathbf{I}$ and $\mathbf{V} = \sigma_g^2\mathbf{H}$. This parameterization assumes a positive genetic scale; the zero-genetic-variance limit reduces to a linear model.

The covariance shape is fitted under the null $\alpha_j = 0$. For fixed δ, define the residual precision matrix

$$\mathbf{P}_H = \mathbf{H}^{-1} - \mathbf{H}^{-1}\mathbf{X}(\mathbf{X}^\top\mathbf{H}^{-1}\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{H}^{-1}. \tag{3}$$

Up to constants independent of δ, the profiled restricted log-likelihood is

$$\ell_R(\delta) = -\frac{1}{2}\left\{\log|\mathbf{H}| + \log|\mathbf{X}^\top\mathbf{H}^{-1}\mathbf{X}| + (n-p)\log\left(\frac{\mathbf{y}^\top\mathbf{P}_H\mathbf{y}}{n-p}\right)\right\}. \tag{4}$$

Thus $\widehat{\sigma}_g^2 = \mathbf{y}^\top\mathbf{P}_H\mathbf{y}/(n-p)$ for REML at the fitted δ. Maximum likelihood instead uses n and omits the covariate determinant term. A phenotype entirely explained by $\mathbf{X}$ has zero residual variance and is rejected rather than assigned numerical noise as a variance estimate.

LOCO matrices can be formed by subtracting group-specific cross-products from the genome-wide sum. The dense implementation refits variance components for each group, by Cholesky factorizations since version 2.0.0.dev5 (Section 2.10); the streaming infinitesimal path uses one global variance fit and group-specific solves. These are distinct nuisance-fitting procedures. When chromosome labels are numerous, the implementation can merge them into a limited number of groups, so the actual grouping must accompany the results. Exclusion reduces direct overlap between the tested region and its background predictor; it does not imply statistical independence across groups or correct an omitted phenotype mean.

2.3 An explicit moment-estimation option for HRATT

Variance estimation can dominate a streaming fit even after individual matrix products are efficient. HRATT already computes randomized Haseman–Elston (HE) quantities to choose the frequency-weight exponent a, with $w_j = \{f_j(1-f_j)\}^{1+a}$ in Equation (1). The optional `heritability_method="he"` reuses these products to estimate the genetic and residual covariance coefficients. The default, `heritability_method="auto"`, keeps REML without sampling weights and uses HE with them (Section 2.12). HE avoids REML's iterative solves and stochastic likelihood calculations, but it is a different estimator.

Let Q span the covariates orthonormally, $S = \mathbf{I} - QQ^\top$, $d = n - p$, and $K = S\mathbf{K}_w S$. After residualizing and scaling the phenotype, write $\mathbf{y}_s = S\mathbf{y}/\sqrt{\mathbf{y}^\top S\mathbf{y}/d}$. The working covariance is $v_g K + v_e S$. The noise term must be S, not $\mathbf{I}$: projection makes its off-diagonal entries nonzero. Frobenius least squares against $\mathbf{y}_s\mathbf{y}_s^\top$ gives the two moment equations

$$\begin{pmatrix} t_2 & t_1 \\ t_1 & d \end{pmatrix} \begin{pmatrix} v_g \\ v_e \end{pmatrix} = \begin{pmatrix} \mathbf{y}_s^\top K \mathbf{y}_s \\ \mathbf{y}_s^\top \mathbf{y}_s \end{pmatrix}, \qquad t_1 = \operatorname{tr} K, \quad t_2 = \operatorname{tr} K^2. \tag{5}$$

Independent Rademacher probes r_b estimate t_2 by $B^{-1}\sum_{b=1}^{B} \|Kr_b\|^2$, with $B = 32$ by default. The diagonal and products already computed for exponent selection supply the remaining terms. All exponents use the same probes. Exponent selection retains the existing off-diagonal HE score; it is not replaced by the constrained fit in Equation (5).

With $D = t_2 - t_1^2/d > 0$, the unconstrained genetic coefficient is $\{\mathbf{y}_s^\top K\mathbf{y}_s - (t_1/d)\mathbf{y}_s^\top\mathbf{y}_s\}/D$. If either fitted coefficient is nonpositive, the implementation compares the two nonnegative boundary solutions of the same quadratic objective. It does not clip the unconstrained ratio to an arbitrary interval. Nonfinite or numerically unidentifiable moments, including nonpositive curvature, raise an error. The returned diagnostics retain the raw and constrained coefficients, boundary status, D, and the Monte Carlo standard error of t_2. This last quantity is not a confidence interval for heritability.

The ratio passed to the marker-effect prior is

$$h^2 = \frac{v_g}{v_g + v_e}, \qquad h^2_{\text{projected}} = \frac{v_g t_1/d}{v_g t_1/d + v_e}. \tag{6}$$

The first follows the package's covariance-coefficient convention; only when $t_1/d = 1$ is it also the mean projected variance fraction. Both are reported. A zero genetic coefficient gives zero polygenic effects. The variational fit retains its existing residual-variance floor of 0.001 on the standardized phenotype scale, including at the opposite boundary; the effective value is reported.

This is a projected single-component moment estimator. LDAK-KVIK instead uses a partitioned HE procedure with large-effect exclusions and an optional Monte Carlo REML revision (Hof and Speed, 2025), which the pinned reference implementation applies depending on sample size and its structure workflow. The two programs' HE fits therefore target different covariance models.

2.4 What an exact scan makes exact

For a fixed covariance shape $\mathbf{H}$, let $U_j = \mathbf{z}_j^\top \mathbf{P}_H \mathbf{y}$, $q_j = \mathbf{z}_j^\top \mathbf{P}_H \mathbf{z}_j$, and $S_0 = \mathbf{y}^\top \mathbf{P}_H \mathbf{y}$. Generalized least squares gives

$$\widehat{\alpha}_j = \frac{U_j}{q_j}, \qquad F_j = \frac{U_j^2/q_j}{(S_0 - U_j^2/q_j)/(n-p-1)}. \tag{7}$$

Under a Gaussian null with this *fixed, correct* covariance shape, $q_j > 0$, and positive residual degrees of freedom, $F_j \sim F_{1,n-p-1}$. In EMMAX-style scans, however, δ is first estimated from the

phenotype and then held fixed while variants are tested (Kang et al., 2010). The same algebra can be computed accurately, but the finite-sample reference law is then a plug-in approximation. Throughout this report, "exact scan" refers to the dense calculation with its fitted covariance, not a guarantee of exact type-I error.

The corresponding known-covariance score test is particularly useful for analysis. Replace $\mathbf{H}$ by the fully specified $\mathbf{V}$ in Equation (3) to obtain $\mathbf{P}_V$. Since $\mathbf{P}_V\mathbf{V}\mathbf{P}_V = \mathbf{P}_V$, the null statistic satisfies

$$T_j = \frac{(\mathbf{z}_j^\top \mathbf{P}_V \mathbf{y})^2}{\mathbf{z}_j^\top \mathbf{P}_V \mathbf{z}_j} \sim \chi^2_1. \tag{8}$$

This conditional result is the reference for the new experiment. It avoids conflating an approximation to the denominator with uncertainty in estimated variance components. Variants collinear with the covariates have zero residual norm and are not estimable.

2.5 Scalar calibration and its spectral meaning

Work in orthonormal coordinates spanning the complement of $\mathbf{X}$. Let $R^\top R = \mathbf{I}_d$, $R^\top \mathbf{X} = 0$, and $d = n - p$; use $\widetilde{\mathbf{y}} = R^\top \mathbf{y}$, $\widetilde{\mathbf{z}}_j = R^\top \mathbf{z}_j$, and $\widetilde{\mathbf{K}} = R^\top \mathbf{K} R$. For the rest of this subsection the tildes are suppressed. The infinitesimal solve is $\mathbf{w} = (\mathbf{K} + \delta \mathbf{I})^{-1}\mathbf{y}$. A retrospective statistic uses $D_j^{\mathrm{ret}} = (\mathbf{z}_j^\top \mathbf{z}_j / d)\|\mathbf{w}\|^2$ in the denominator of $(\mathbf{z}_j^\top \mathbf{w})^2$. The matching prospective denominator is $D_j^{\mathrm{pro}} = \sigma_g^2 q_j$, where $q_j = \mathbf{z}_j^\top (\mathbf{K} + \delta \mathbf{I})^{-1} \mathbf{z}_j$. A calibration factor averages $D_j^{\mathrm{pro}} / D_j^{\mathrm{ret}}$ over selected markers. A common factor can reproduce the prospective statistic for all markers only when the relevant ratios are constant.

Write $\mathbf{K} = \mathbf{U}\Lambda\mathbf{U}^\top$ and $a_{\ell j} = \mathbf{U}_\ell^\top \mathbf{z}_j$. The missing geometry is explicit:

$$q_j = \sum_{\ell=1}^{d} \frac{a_{\ell j}^2}{\lambda_\ell + \delta}, \qquad \|\mathbf{z}_j\|^2 = \sum_{\ell=1}^{d} a_{\ell j}^2. \tag{9}$$

A direction assigned high background variance receives low precision. Two variants with equal norms can therefore have different q_j. A denominator calibrated to the average may be too small for variants concentrated in low-variance directions and too large for variants concentrated in high-variance directions.

For known covariance and a deterministic approximate denominator $\widehat{q}_j$, put $r_j = q_j / \widehat{q}_j$. The consequence is exact:

$$\widetilde{T}_j = r_j T_j, \qquad \Pr(\widetilde{T}_j > c_\alpha \mid \mathbf{Z}) = \overline{F}_{\chi^2_1}\left(\frac{c_\alpha}{r_j}\right), \quad c_\alpha = F^{-1}_{\chi^2_1}(1-\alpha). \tag{10}$$

An error that appears modest near the center can matter much more in the tail. Averaging Equation (10) over a variant group gives its expected rejection fraction, without assuming variants are independent. It is a marginal rejection probability, not the family-wise probability of at least one discovery.

2.6 A low-rank correction and its current limits

Retaining the leading k eigenpairs and replacing the rest by their mean $\bar{\lambda}_{>k}$ gives the approximation

$$q_j^{(k)} = \sum_{\ell \le k} \frac{a_{\ell j}^2}{\lambda_\ell + \delta} + \frac{\|\mathbf{z}_j\|^2 - \sum_{\ell \le k} a_{\ell j}^2}{\bar{\lambda}_{>k} + \delta}, \qquad \widehat{q}_j = c_k q_j^{(k)}. \tag{11}$$

The scalar c_k corrects average remaining error; it cannot remove arbitrary variation in the spectral tail. For a LOCO analysis, each approximation should target the covariance used by that group's test. Using a full-genome basis to approximate a different LOCO covariance requires its own error assessment.

The production implementation computes group-specific bases with randomized subspace iteration and currently tries widths 64, 128, 256, and 512. It stops when the coefficient of variation of calibration ratios is at most 3%, subject to the available dimension. Because the calibration markers also set the scale, this stopping rule is an internal diagnostic; it bounds no error at genome-wide thresholds. The production infinitesimal calibration also selects candidate markers using a phenotype-dependent statistic; this differs from the phenotype-independent selection in our controlled experiment.

For mixture-prior and elastic-net residuals, an infinitesimal spectral factor is transferred before further calibration. Once the residual is a nonlinear, data-adaptive function of $\mathbf{y}$, the proof of Equation (8) no longer supplies its null law, and the transfer is a heuristic. "BOLT-style" and HRATT below denote mixmogam methods inspired by BOLT-LMM and LDAK-KVIK, not the reference programs.

2.7 Environmental mean misspecification

Suppose the actual conditional mean includes an omitted exposure vector $\mathbf{d}$ with effect γ: $\mathbb{E}(\mathbf{y} \mid \mathbf{Z}, \mathbf{d}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{d}\gamma$. Even if the covariance $\mathbf{V}$ is known correctly, the numerator in Equation (8) then has mean $\mathbf{z}_j^\top \mathbf{P}_V \mathbf{d}\gamma$. Its distribution becomes

$$T_j \sim \chi^2_1(\eta_j), \qquad \eta_j = \frac{(\mathbf{z}_j^\top \mathbf{P}_V \mathbf{d}\gamma)^2}{\mathbf{z}_j^\top \mathbf{P}_V \mathbf{z}_j}. \tag{12}$$

A kinship matrix can assign large variance to an ancestry direction, but it does not automatically place an environmental mean shift in $\mathbf{X}$. Adding covariates that span $\mathbf{d}$ makes the numerator mean zero under this model. Estimated PCs may only approximate that span, and neither the required number of PCs nor a power-preserving adjustment follows from this identity alone. Reviews of structure and mixed models place this distinction in a broader setting (Price et al., 2010; Sul et al., 2018; Yang et al., 2014).

2.8 Numerical implementation and ancillary inference

Dense scans whiten genotype blocks and use matrix products to evaluate Equation (7). Dense factorization costs $O(n^3)$ time and $O(n^2)$ storage, with a dense transformation cost of order $n^2 m$ for m tested variants. Streaming methods replace stored kinships by products with standardized genotype blocks. A product with r right-hand sides costs order nMr arithmetic; its memory consists of the stored calls, the right-hand sides and working blocks; the calls alone take one byte, or a quarter byte, per genotype.

Conjugate gradients supplies covariance solves, and stochastic Lanczos quadrature estimates likelihood terms for large samples. Both introduce numerical approximations beyond the statistical model. Association solves now reject nonconvergence. Variational fits expose incomplete convergence, and fit caches respect their options and model ownership. The review also removed an unnecessary LOCO-group axis from temporary products; Section 4.5 reports the measured allocation change with its scope.

Several adjacent routines need equally explicit contracts. In the genetic BLUP (Equation (22)) the genetic scale cancels and must not be multiplied in a second time. Default permutation inference operates on orthogonal residual coordinates rather than permuting the correlated entries of a projected residual vector, following the underlying distinction developed by Abney (2015). Exact exchangeability requires the stipulated Gaussian covariance; substituting fitted variance components still requires calibration checks. Finally, the normal-approximation Bayes factor requires effect estimates, standard errors, and an explicit effect-variance prior (Wakefield, 2009); it does not replace a joint fine-mapping model. Appendix A gives these identities and their interpretation.

2.9 Reuse, bounded workspaces and optional parallelism

The computational changes first remove work whose result is already available. Variational fitting caches requested fold-specific Gram matrices as packed triangles in the genotype storage precision, block by block under a shared 10^9-byte default budget. HRATT's cross-fit cache can expand after the hold-out cache is released. It reuses full-block Gram matrices for LOCO and reuses final predictions. A persistent matrix-product workspace replaces repeated allocations. Residual updates retain float64 accumulation while preparing the next matrix-product input in the genotype storage precision, float32 by default. Converting the entire calculation to float32 would be a different numerical contract.

Temporary widening is bounded without dividing the reduction axis: compiled genotype preparation keeps a few float64 values per marker of a 64-marker tile (the NumPy fallback uses tiles with a 64 MiB scratch target); prediction, HE diagonal terms and retrospective statistics use 16 MiB targets. These are explicit temporary-workspace budgets, not process RSS limits. Genotype validation and PLINK reading also use bounded chunks; reading no longer retains an entire packed BED beside its decoded copy.

With Numba, `n_threads>1` distributes the preparation's 64-marker tiles across workers; each marker's samples are still summed in order, so the prepared values do not depend on the thread count (Section 2.11). Retaining Mp float64 projection coefficients permits every later pass to decode directly from the stored calls in bounded slices, which bound the decoding workspace, not total memory.

Coordinate sweeps parallelize independent candidate models and LOCO fits while preserving the Gauss–Seidel marker order within each model. In multi-model fits from 50,000 samples, residual matrix products also use an output-row partition and a pool capped at four workers; 2.0.0.dev6 also required six models (Section 4.8). Each output entry retains its complete marker reduction; there is no shared writable sum across workers. Detected BLAS pools are temporarily limited to one, and thread settings and workers are restored on exit. Apple Accelerate requires an external thread limit because it is not detected by `threadpoolctl`. Parallel residual products and BLAS reductions can still change floating-point rounding.

2.10 Computing the same estimators with less work

The changes below, introduced in 2.0.0.dev5, compute the quantities of the preceding subsections with less work and leave the models and estimators unchanged. Three of them do change results. Two-step fits draw different trace probes and calibration markers for a given seed. The Lanczos defaults trade quadrature steps for probes. MLMM forward scans default to float32. Section 4.9 reports the measured effects.

Exact LOCO REML by Cholesky factorization. Equation (4) needs $\log|\mathbf{H}|$, $\log|\mathbf{X}^\top\mathbf{H}^{-1}\mathbf{X}|$ and $\mathbf{y}^\top\mathbf{P}_H\mathbf{y}$, not the spectrum of $\mathbf{K}_{-g}$. With $\mathbf{H} = \mathbf{L}\mathbf{L}^\top$, $\log|\mathbf{H}| = 2\sum_i \log L_{ii}$, and a least-squares fit of $\mathbf{L}^{-1}\mathbf{y}$ on $\mathbf{L}^{-1}\mathbf{X}$ supplies the other two terms. A likelihood evaluation is therefore one factorization, about $n^3/3$ operations in a parallel BLAS-3 kernel. The eigendecomposition it replaces used about 1.2 cores of the test machine: at $n = 4{,}000$ it took 6.4 seconds against 0.12 seconds for a factorization. The search over $\log\delta$ keeps the EMMA grid limits $[-10, 10]$. For the first group, a 21-point grid locates the maximum and bounded Brent refinement resolves it to an absolute 10^{-6}. Later groups, whose kinships share most markers, expand a bracket from the previous optimum. If the first profile has more than one interior maximum, every group repeats the grid.

The F statistic of Equation (7) depends on $\mathbf{z}_j$, $\mathbf{y}$ and $\mathbf{X}$ only through inner products in the metric $\mathbf{H}^{-1}$. Any whitening $\mathbf{W}$ with $\mathbf{W}^\top\mathbf{W} = \mathbf{H}^{-1}$ gives the same statistics, so the LOCO scan whitens genotype blocks by triangular solves with $\mathbf{L}$ at the fitted δ. Where $\mathbf{K} = \mathbf{U}\Lambda\mathbf{U}^\top$ is computed anyway, as in non-LOCO scans and MLMM, $\mathbf{W} = (\Lambda + \delta\mathbf{I})^{-1/2}\mathbf{U}^\top$ takes one matrix product per block, against two for the symmetric root $\mathbf{U}(\Lambda+\delta\mathbf{I})^{-1/2}\mathbf{U}^\top$. Permutation routines permute sample entries and keep the symmetric root, so a seed reproduces its earlier results. MLMM refits δ as cofactors enter, but $\mathbf{U}$ stays fixed. The rotated genotypes $\mathbf{U}^\top\mathbf{Z}$ are therefore formed once, within a memory budget. Each forward scan then rescales them and removes the cofactor directions: $O(nMq)$ operations for q cofactors rather than $O(n^2M)$ per step.

Lanczos steps and probes. Stochastic Lanczos quadrature evaluates the REML objective from Lanczos runs on the deflated operator. A run from the projected phenotype gives the quadratic form; runs from N Rademacher probes give the log-determinant and trace terms. One quadrature rule per run serves every δ. A fit therefore costs s operator products with $N + 1$ columns for s Lanczos steps, and each product is one pass over the genotypes. Quadrature error falls quickly with s for these smooth functions of a positive definite operator, whereas probe error falls only as $N^{-1/2}$. On the test machine, a pass cost nearly the same for 1 and 64 columns, because decoding and memory traffic dominate. The defaults are therefore $s = 24$ and $N = 48$ (96 and 12 before 2.0.0.dev5): a quarter of the passes and four times the probes. Here 24 is the cap of an adaptive stop controlled by `slq_tol`, so fewer steps may be used. Probes that the covariate projection reduces to rounding noise are discarded.

Shared spectral basis and calibration solves. Streaming genotype blocks are stored after covariate projection. The streaming kinship therefore already equals the REML operator $\mathbf{SKS}$, with $\mathbf{S} = \mathbf{I} - \mathbf{Q}\mathbf{Q}^\top$ for an orthonormal covariate basis $\mathbf{Q}$. In the BOLT-style paths, and in HRATT under strong structure, one randomized subspace iteration of width 128 supplies both the REML deflation and the 64-pair conjugate-gradient preconditioner; previously each ran its own passes. The infinitesimal calibration of Section 2.6 uses random markers whose retrospective statistic is below 5. That statistic depends on residuals that do not yet exist when their solves start. Polymorphic candidates are therefore drawn in random order beforehand, with 25% oversampling, and solved in the residuals' batched conjugate-gradient run. The first n_{cal} that qualify are kept, and a short second solve covers any shortfall. This rejection sampling still selects a uniformly random subset of the qualifying markers, but a given seed selects different markers than before.

Streamed genotypes. With its float cache disabled, 2.0.0.dev5 rebuilt standardized, projected blocks from int8 calls through per-variant value tables, matching the cached blocks bit for bit;

Section 2.11 describes the current streaming. With Numba, dense relationship matrices also standardize each genotype block in one compiled pass.

2.11 Genotype streaming without per-marker projection

Genotype streaming keeps the estimators and moves the covariate projection off the markers. With $\mathbf{z}_j$ a marker's standardized, unprojected values and $\mathbf{Q}$ the orthonormal covariate basis, preparation stores the three standardized call values, the coefficients $\mathbf{c}_j = \mathbf{Q}^\top \mathbf{z}_j$ and the squared norm $\|\mathbf{S}\mathbf{z}_j\|^2 = \|\mathbf{z}_j\|^2 - \|\mathbf{c}_j\|^2$. One compiled kernel computes them in float64 over tiles of 64 markers, reading one sample at a time and summing each marker's samples in order. No projected block is formed. Later passes decode unprojected values by table lookup (missing calls decode to zero) and remove the covariates where that is cheapest. With $\mathbf{Z}_b$ the standardized, unprojected columns of block b, an $n \times c$ operand $\mathbf{B}$ and the result are each projected once:

$$\mathbf{KB} = \frac{1}{M}\,\mathbf{S}\Big(\sum_b \mathbf{Z}_b\mathbf{Z}_b^\top\Big)(\mathbf{SB}). \tag{13}$$

The variational sweep corrects each 128-marker subblock's products, $\mathbf{Z}_b^\top\mathbf{S}\mathbf{r} = \mathbf{Z}_b^\top\mathbf{r} - \mathbf{C}_b^\top(\mathbf{Q}^\top\mathbf{r})$ with $\mathbf{C}_b = \mathbf{Q}^\top\mathbf{Z}_b$, and its residual updates alike. Subblock Gram matrices, the HE diagonal and LD scores need products of projected markers, which are unprojected products plus a rank-q correction,

$$(\mathbf{S}\mathbf{z}_j)^\top(\mathbf{S}\mathbf{z}_k) = \mathbf{z}_j^\top\mathbf{z}_k - \mathbf{c}_j^\top\mathbf{c}_k. \tag{14}$$

Calls are stored two bits each, as PLINK BED rows, in memory or memory-mapped, and decoded directly (commit e4089d8, which also removed an optional float cache that in commit 12aa504 held the same unprojected values, with bit-identical cached and streamed fits); preparation takes each marker's mean and variance from exact call counts, so int8 and two-bit storage, every layout and every thread count prepare identical values. Decoding follows the storage order: on one core of the test machine it costs about 0.2 ns per genotype from two-bit calls, 0.26 ns from marker-major int8 calls and 0.7 ns from sample-major arrays. One-pass consumers decode 16 MiB slices, which at 10,000 samples stay in cache between decoding and their two products; a floor of 256 markers per slice, up to 512 MiB, amortizes the per-slice $n \times c$ accumulation at larger n. Slices and the 128-marker variational subblocks reuse their buffers: a fresh allocation per block faulted in every page and doubled the decoding time. BOLT-LMM's in-sample LD scores stream each chromosome through a sliding position window and hold only its widest window, rather than every projected genotype at four bytes each.

2.12 Case-control outcomes and sampling weights

HRATT also analyses binary outcomes (`trait="binary"`) and samples with sampling weights (`sample_weights`), such as inverse probabilities of participation (Schoeler et al., 2023), alone or together. Its score tests carry a calibration factor c: one, unless LDAK-KVIK's test of inter-chromosome correlation finds strong structure, when c matches the statistics to GRAMMAR-Gamma-calibrated ridge statistics on weakly associated markers (Hof and Speed, 2025).

Let ω be the sampling weights scaled to mean one and ν the working weights: $\nu = \omega$ for a quantitative trait and $\nu = \omega\,\mu_0(1-\mu_0)$ for a binary one, where μ_0 is the covariate-only (weighted) logistic fit. The working response $\mathbf{y}^*$ is $\mathbf{y}$, or $(y_i - \mu_{0i})/\{\mu_{0i}(1-\mu_{0i})\}$ for a binary trait. With $s = \sqrt{\nu}$, step 1 fits $\tilde{\mathbf{y}} = s \circ \mathbf{y}^*$ on $\tilde{\mathbf{X}} = s \circ \mathbf{X}$ and genotypes $\tilde{\mathbf{z}}_j = s \circ \mathbf{z}_j$: weighted least squares that keeps every solver's form. For a binary trait this is one iteratively reweighted least-squares step,

as LDAK-KVIK's binary step 1 is a weighted linear regression (Hof and Speed, 2025). Constant ν leaves step 1 unscaled.

Without sampling weights REML fits the working model, so a binary trait's h^2 is on its scale. REML has no likelihood under sampling weights, so weighted fits use HE, and requesting REML for the heritability or the frequency-weight exponent is an error. With $\tilde{S}$ the projector off $\tilde{\mathbf{X}}$ and $D_\omega = \mathrm{diag}(\omega)$, the noise covariance of the scaled model is $\tilde{S}D_\omega\tilde{S}$; the least-squares fit of $\tilde{\mathbf{y}}_s\tilde{\mathbf{y}}_s^\top$ on $v_gK + v_e\tilde{S}D_\omega\tilde{S}$ weights each diagonal entry by $1/\omega_i$, so that every moment is design-consistent. With unit weights its normal equations are Equation (5). The structure test uses the Kish size of ν in place of n.

Step 2 tests the score $U_j = \mathbf{z}_j^\top\mathbf{a}_g$ of variant j on chromosome group g, where $\mathbf{a}_g = s \circ \tilde{\mathbf{r}}_g$ for a quantitative trait, $\tilde{\mathbf{r}}_g$ being the scaled LOCO residual projected off $\tilde{\mathbf{X}}$, and $\mathbf{a}_g = \omega \circ (\mathbf{y} - \boldsymbol{\mu}_g)$ for a binary one, $\boldsymbol{\mu}_g$ being the weighted null logistic fit with the logit-scale LOCO score as offset. Both are orthogonal to the unscaled covariates. When genotypes are exchangeable given the covariates, whatever the weights,

$$\mathrm{Var}(U_j \mid \mathbf{a}_g) \approx \zeta_j \,\|\mathbf{a}_g\|^2, \qquad \zeta_j = \frac{\mathbf{z}_j^\top\mathbf{z}_j - \|Q_X^\top\mathbf{z}_j\|^2}{n-p}, \tag{15}$$

where ζ_j, with Q_X an orthonormal basis of the unscaled covariates, estimates the genotype variance given the covariates. For an unweighted quantitative trait, $U_j^2/\{\zeta_j\|\mathbf{a}_g\|^2\}$ is the unweighted statistic, and that analysis is unchanged, with normal tails. In weighted and binary analyses a saddlepoint approximation replaces the normal tail above $|z| = 2$ (`spa_threshold`). It treats the genotypes as independent draws from the variant's empirical distribution of values (at most four, counting the no-call value), with $\mathbf{a}_g$ fixed: the retrospective counterpart of SPACox's empirical-CGF approximation, which randomizes the residuals instead (Bi et al., 2020). Its shape comes from the marginal genotype distribution, not adjusted for covariates, and its scale from Equation (15). The factor c is computed on the working model's linear statistic and multiplies the retrospective one, a heuristic transfer. Quantitative effects are weighted least-squares slopes; binary ones are one-step log odds ratios U_j/J_j, with J_j the logistic information of the covariate-adjusted genotype. Separation by the covariates is an error, while per-group null fits that do not converge are reported rather than rejected. The spectral denominator is unavailable with weights or binary traits, and the other association methods reject these options. Unit sampling weights still select HE and these tests.

Two common alternatives failed in pilots and are not used. The Huber–White sandwich $\sum_i a_{gi}^2 x_{ij}^2$, with x_{ij} the weighted covariate-adjusted genotype, replaces each genotype's variance by its own square. Under selection on the outcome a few samples carry each score: its test was inflated 17-fold at $\alpha = 10^{-3}$ for minor allele frequency 1–5%, where Equation (15) with the saddlepoint gave 0.75-fold. The model-based logistic variance $\sum_i \mu_i(1-\mu_i)\tilde{g}_{ij}^2$ of logistic score tests with saddlepoint tails (Dey et al., 2017; Zhou et al., 2018; Hof and Speed, 2025) trusts fitted probabilities that an in-sample LOCO offset overfits when cases are few: with HRATT's offsets and 50 cases among 5,000 it gave $\lambda_{\mathrm{GC}} = 0.55$ on a null trait, against 1.01–1.04 for Equation (15).

2.13 Cross-fitted scores and covariate-specific genotype distributions

The validation of Section 4.11 found three faults. HRATT's effects were attenuated, rare binary outcomes broke the per-group null fits, and weights that depend on ancestry inflated low-frequency tails. The changes below address them. Section 4.12 reports the prespecified validation of these changes; the figures quoted here come from prototypes.

In-sample absorption. A LOCO score fitted on all samples carries part of each sample's own phenotype. For a linear smoother S_g of the other chromosomes, the residual $\mathbf{r}_g = (\mathbf{I} - S_g)\mathbf{y}$ of a variant with effect β_j on chromosome group g has

$$\mathbb{E}(\mathbf{z}_j^\top \mathbf{r}_g) = \mathbf{z}_j^\top (\mathbf{I} - S_g)\mathbf{z}_j \, \beta_j \approx (1 - \operatorname{tr} S_g/n)\, \mathbf{z}_j^\top \mathbf{z}_j \, \beta_j,$$

so the slope $\mathbf{z}_j^\top \mathbf{r}_g / \mathbf{z}_j^\top \mathbf{z}_j$ is attenuated by about $\operatorname{tr} S_g/n$ (the bias GRAMMAR-Gamma corrects; Svishcheva et al., 2012). The residual variance shrinks with it, so p-values are unaffected. For a binary trait the working response of a case is about $1/\mu_0$, and the same absorption gives the cases offsets of tens of logit units at 1% prevalence: on one null replicate cases averaged +43 against −0.4 for controls. Heritability on the Pearson scale was overestimated there (REML median 0.10 on null traits at 1% prevalence, up to 1.0), which made it worse.

Cross-fitted scores. HRATT follows REGENIE's level 0 (Mbatchou et al., 2021):

- The samples are split into k folds (`loco_folds`, default 5), by case status for binary traits.
- The selected elastic-net model is fitted genome-wide on all but one fold, k fits in one sweep.
- A sample's score for group g is its own fold's fit without group g's variants. With strong structure, each group's model is refitted without its variants instead (k fits per group).

Each fold's scores are then trained without that fold's outcomes. The variance components and the prior are still chosen on all samples, so the procedure as a whole does not exclude a sample's outcome from its own score. Each group's score enters with its least-squares coefficient, or logistic coefficient for binary traits, kept in $[0, 1]$. This out-of-fold calibration slope shrinks an over-dispersed score but never undoes the prior's shrinkage. Binary scores with a coefficient above one become fixed offsets. Scores that anti-predict, or whose null fit diverges, are dropped for their group. Without polygenic variance there are no scores.

LDAK-KVIK instead keeps in-sample scores and calibrates effects by the slope of effects estimated without the score on effects estimated against the residual. That reference is confounded under structure: at $F_{ST} = 0.05$ it gave 1.24 for minor allele frequency 1–5% and 1.37 above, and calibrated QTL effects came out 15% large. It also cannot repair binary offsets.

Without structure, dropping a group's share of a genome-wide fit matched per-group refits in prototypes: null λ_{GC} was 0.975–1.023 against 0.975–1.018, with the same QTL effects and power. It costs k fits instead of k per group. Under structure it fails. A genome-wide fit spreads the ancestry signal over every chromosome, so dropping one chromosome's share leaves ancestry in its residual. At $F_{ST} = 0.15$ without principal components, the null chromosome's λ_{GC} was 1.5–4.4, against 0.9–1.1 refitted. HRATT refits each group when the structure test finds strong structure. Below the test's threshold of 0.1 in average inflation, dropping one of G chromosomes' shares leaves roughly $1/G^2$ of it.

Covariate-specific genotype distributions. Equation (15) treats the genotypes as exchangeable given the covariates. When allele frequencies vary with ancestry and the weights or case fractions do too, that fails. Following SPAmix (Ma et al., 2025), sample i's allele frequency p_{ij} is half the genotype count fitted on the covariates. Its deviations from the mean are shrunk by $(F - 1)/F$, with F the regression's F statistic, so that covariates unrelated to the genotype leave the pooled frequency. With $\pi_{ij} = p_{ij}(1 - p_{ij})$,

$$\operatorname{Var}(U_j \mid \mathbf{a}_g) \approx \zeta_j \, \|\mathbf{a}_g\|^2 \rho_j, \qquad \rho_j = \frac{\sum_i a_{gi}^2 \pi_{ij} / \sum_i a_{gi}^2}{n^{-1} \sum_i \pi_{ij}}, \tag{16}$$

which is Equation (15) when the covariates do not predict the genotype. Ancestry principal components must therefore be among the covariates. HRATT warns when the structure test still finds strong structure after the covariates. The package computes them (`mixmogam.pca`) as the leading left singular vectors of a thinned panel: by default 2,000 random markers with sample minor allele frequency at least 5%, standardized, by an exact float64 SVD with fixed signs. A conservative working-memory estimate is checked before scanning frequencies or allocating the panel, with a configurable default budget of 4 GiB. This excludes resident input calls and library overhead; it is an allocation guard rather than a measured peak-memory bound.

The saddlepoint uses the same model. Each called genotype is an independent Binomial$(2, p_{ij})$ draw given the covariates and the observed missing calls, which assumes Hardy–Weinberg equilibrium given the covariates. An intercept-only design keeps the empirical genotype distribution. The pooled distribution with only its variance corrected, used before 2.0.0.dev6, over-rejected 4.7-fold at 10^{-4} against an exact convolution of ancestry-specific binomials.

Two-sided tails. `spa_two_sided="doubled"` doubles the saddlepoint tail beyond the observed score, LDAK's default, putting $\alpha/2$ in each tail of a skewed null. The default still adds both tails at $\pm|u|$, as SPAtest, SAIGE and REGENIE do (Dey et al., 2017; Zhou et al., 2018; Mbatchou et al., 2021).

Weighted and binary standard errors describe the null-score linearisation and do not depend on the tail rule. Binary effects remain one-step approximations; exports label them `beta_one_step` and `se_null_score`.

3 Evaluation design and provenance

3.1 New conditional experiment with known covariance

The new experiment isolates Equation (10); it does not call an end-to-end GWAS scan. We generate 256 diploid samples in four equal populations, with 2,000 background and 1,000 candidate markers. For each marker, the ancestral frequency f_j is uniform on $[0.1, 0.9]$. Population-specific frequencies are independent draws from Beta$(f_j(1/\theta-1), (1-f_j)(1/\theta-1))$ for $\theta > 0$; at $\theta = 0$ they all equal f_j. Hard calls are binomial with two trials. The values $\theta \in \{0, 0.05, 0.20\}$ control frequency dispersion. We do not report them as measured F_{ST}, whose estimation and interpretation require a separate definition (Bhatia et al., 2013).

Markers are independent conditional on population frequencies and sample membership. Population structure nevertheless correlates their sample-level patterns. Monomorphic markers are removed: one background marker is dropped in the $\theta = 0.20$ panel; all 1,000 candidates remain in every panel. The package's raw relationship matrix is checked against a direct standardized cross-product. We remove an intercept in orthonormal residual coordinates, set $\delta = 1$ and genetic scale one, and use a full eigendecomposition to obtain the known covariance. Direct linear solves and spectral denominators agree to numerical precision.

Thirty randomly selected candidates calibrate each approximation; the other 970 are held out for audit. Selection is independent of phenotype. The scalar denominator is the variant norm times the mean $q_j/\|\mathbf{z}_j\|^2$ among calibration markers. For the spectral method, we try $k \in \{4, 8, 16, 32, 64, 128\}$ and select the first width whose calibration-ratio coefficient of variation is at most 3%. All three panels select $k = 4$. Exact leading eigenvectors make this a favourable test of spectral geometry; the production randomized eigensolver is not exercised.

Audit variants are divided into quintiles by the share of their squared norm explained by the top three background eigenvectors. For each method and quintile we calculate the analytic mean

rejection probability at $\alpha = 0.01$ and 5×10^{-8}. We also simulate 2,000 independent Gaussian phenotype replicates per fixed panel and compare the empirical rejection fraction at 1% with its analytic value. Methods use the same phenotypes. Monte Carlo standard errors are calculated from the replicate-level fractions, retaining correlation among variants within each replicate. We do not estimate the extreme-tail probability by counting a handful of simulated exceedances.

The three genotype panels use seeds 20261003–20261005. Consequently, differences across θ also include differences among three realized panels; there are no genotype-replicate confidence intervals. The analytic result is exact conditional on each realized panel and its fixed approximation.

3.2 Matched phensim benchmark with official LDAK-KVIK

We froze a factorial protocol before examining association outcomes and used an excluded pilot to check interoperability and resource demands. Each cell has six independent genotype and phenotype replicates, $n = 800$ diploid samples, and $m = 6{,}000$ markers on six equal chromosomes. All genotypes and phenotypes are generated by phensim. Its `simulate_population_structure` function uses three populations with uniformly sampled membership and Balding–Nichols frequency drift. The software's `fst` parameter is zero or 0.05; it controls frequency dispersion and is not an empirical estimate of F_{ST}. Shared ancestral frequencies are $0.3 + N(0, 0.05^2)$, clipped to $[0.05, 0.95]$; population frequencies are clipped to $[0.01, 0.99]$.

Two LD settings isolate local linkage from ancestry correlation. Markers are either independent conditional on population or generated in 50-marker blocks using two latent Gaussian AR(1) haplotypes per person with correlation parameter 0.8. Population-specific thresholds convert each haplotype to allele indicators. Blocks are independent conditional on population, but pooled genotypes share ancestry correlations across blocks and chromosomes. We added the optional block generator within phensim, preserving its existing independent-marker seeded draws. Markers with pooled minor-allele frequency below 0.01 are removed before either program receives them; calls are complete.

The base global-null trait has no genetic effects and Gaussian residuals. The mixed trait has target genetic variance 0.5, divided equally between an infinitesimal background and ten normally distributed QTL effects. Both components use chromosomes 1–5 only, with QTLs in distinct blocks; chromosome 6 has no generating effects. Shared effects across populations are assumed. Phensim records every component, causal index and effect. For environmental confounding we pass population labels $-1, 0, 1$ as an explicit exposure E, standardized within the sample, and form

$$y = \text{standardize}\left\{\sqrt{s}E + \sqrt{1-s}(u + q + e)\right\}, \qquad s \in \{0, 0.2\}. \tag{17}$$

The component targets need not equal realized variance fractions: population structure can correlate E, u and q. The archive therefore includes their covariance matrix.

Four analysis cells comprise unstructured data, structured data without environmental effects, structured data with the exposure, and the same confounded phenotype adjusted for the two leading genotype PCs. PCs are computed without phenotypes and supplied identically to all methods. Every analysis includes an intercept. Exact LOCO EMMAX, mixmogam BOLT-inf, mixmogam HRATT and official LDAK-KVIK use their default fitting and calibration settings with explicit seeds and one thread. We compare the procedures as supplied; their priors and tuning searches differ. "LDAK-KVIK" denotes the official standalone executable and "mixmogam HRATT" this package's method; they differ in statistical choices, such as variance-component estimation, as well as implementation language. Versions up to commit e4089d8 called the method KVIK, and archived runs keep that name. The official documentation states that LDAK-KVIK has primarily

been tested above 50,000 samples, while also supporting smaller datasets with potentially less benefit from LOCO prediction.[1] The primary panels are therefore small-sample stress tests.

Phensim writes PLINK calls counting BIM allele A2. An independent decoder checks every BED call against the simulated matrix; mixmogam must read the complementary A1 counts. LDAK's own statistics command verifies the same marker IDs, sample IDs, allele labels, allele means and call rates. Both analysis programs read the same BED/BIM/FAM files and serialized phenotypes and covariates. The official native Mac executable `ldak6.3.mac` is pinned to repository revision `995e18753a0a`, with its full URL and SHA256 in the archive; the generic "Version 6" banner alone does not identify this build.

Calibration is measured on all markers under the global genetic null, and separately on chromosome 6 under the mixed trait. The latter shares ancestry with causal chromosomes, so rejection assesses control of ancestry associations as well as statistical approximation. Other markers are not automatically labelled null simply because they are not large QTLs. We report rejection at 0.05, 0.01 and 0.001, genomic inflation from p-equivalent chi-square statistics, and population-loading, frequency and local-LD quintiles. Loading is the fraction of genotype sum of squares explained by population membership, chosen without phenotype outcomes. Power is direct causal-marker detection at $0.05/m$ and 5×10^{-8}. Monte Carlo standard errors use independent replicate panels, not the number of correlated markers.

Each method runs in a separate process. Wall time includes startup, input loading, fitting, scanning and result writing; LDAK-KVIK time sums its two stages. Kernel `wait4` counters provide peak resident memory, using the larger stage peak for LDAK-KVIK. Common simulation, export validation and PC construction are recorded separately. Method order rotates across replicates, code caches may be warm, and the macOS power guard requires AC power with Low Power Mode disabled.

3.3 Larger workloads and reference-based simulation

After inspecting the primary study we fixed a descriptive runtime extension at n=2,000 with m=12,000 and n=4,000 with m=24,000, using two independent replicates for the unstructured and PC-adjusted confounded mixed traits. All four methods retain their fitting defaults. These panels are not pooled into the six-replicate calibration experiment.

For larger n, we implemented the genotype-copying model of HAPNEST (Wharrie et al., 2023) in phensim. A phased reference is expanded by copying segments whose ages and lengths satisfy

$$T \sim \text{Gamma}(2, \text{scale} = N_e/N_{\text{ref}}), \qquad L \mid T \sim \text{Exp}(\text{rate} = 2T\rho). \tag{18}$$

Here N_{ref} counts reference diploids, L is in centimorgans, and T is in generations. Donors are uniform within the assigned population; an allele is retained only if its mutation age exceeds T. Chromosome boundaries restart copying. The independently written implementation follows upstream phase and endpoint conventions, with a fused Numba kernel and a NumPy oracle. It does not implement HAPNEST's ABC parameter tuning, demographic admixture, or upstream phenotype model.

The simulator resource experiment uses 600 phensim reference diploids, 12,000 markers, three frequency-drift populations, latent AR(1) correlation 0.8, map spacing 0.001 cM, mutation ages 1,000 generations, $N_e = 10,000$ and $\rho = 0.7$, chosen as controlled inputs. Three seeded repeats compare backends at 2,000 samples and assess Numba through 100,000 samples. At 50,000 samples, output is streamed, retained in RAM, or memory-mapped. Matched modes must give identical

[1] LDAK-KVIK recommendations, accessed 3 October 2026.

output hashes. First-call compilation or cache loading is timed separately. Warm generation includes batch checksums; both OS process peak RSS and a 2 ms sampled peak are recorded. Memory-mapped file pages can become resident: only streaming avoids materializing the entire result.

The association extension uses n=10,000 and 50,000, m=12,000, and one mixed-trait realization per unstructured or PC-adjusted confounded cell. The unstructured control pools reference donors; the structured cell restricts donors by population. Setting $N_e = 50N_{\text{ref}}$ matches their segment-age law. We retain the primary chromosome-6 genetic null. Tiled phenotype generation preserves the dense GRM factor and Gaussian innovations up to summation roundoff. Genotype PCs use an iterative eigensolver with audited residuals, never an $n \times n$ matrix. Separate preparation processes keep simulation arrays outside association RSS. With one realization per cell, these runs measure feasibility and cost.

3.4 Separating estimator validation from the 20,000-marker workload

We evaluated projected HE before using it for the larger workload. Four independent phensim genotype panels contain $n = 800$, $M = 600$: two unstructured panels and two with frequency-drift parameter 0.08 and two genotype PCs. For each covariance-coefficient target $h^2 \in \{0, 0.3, 0.6\}$, 24 Gaussian phenotypes per panel give 288 draws. The primary comparison fixes $a = -1$ to match the generating covariance and compares 32-probe HE with an independent dense solution of Equation (5). Components are not rescaled to force their realized sample variances. A separate 64-fit experiment compares complete HE and REML scans on the first eight global-null phenotypes per panel.

The 20,000-marker extension uses two new HAPNEST-model panels at $n = 50{,}000$: unstructured, and structured with the environmental exposure in Equation (17) plus two genotype PCs. It retains the six-chromosome, ten-QTL and polygenic-background design of Section 3.3, with 600 simulated reference diploids, AR(1) parameter 0.8, pooled or population-restricted copying, and $N_e = 50N_{\text{ref}}$. From 20,400 candidates, a deterministic genotype-only rule retains the first MAF-passing markers up to chromosome quotas $(3350, 3350, 3350, 3350, 3300, 3300)$. Selection precedes PCs, QTL selection and phenotype generation; original candidate indices are retained. Both panels have zero MAF exclusions and 400 quota removals. Chromosome 6 therefore contains 3,300 genetic-null markers. Six groups preserve the earlier LOCO workload structure, although these are new draws rather than nested subsets of the 12,000-marker panels.

An independent export audit checks every call, allele orientation and sample/marker order; reference-program statistics check the same inputs. PCA uses bounded-memory products and audited eigenpair residuals. All component covariances are retained, since the nominal trait-component targets need not equal realized variance fractions. Common preparation and validation run separately from association processes.

The prespecified primary design contains 24 complete fits: two methods, two requested thread counts (one and four), three timing repetitions and two fixed panels. A separate 12-fit experiment with four mixmogam workers, prespecified to compare cache budgets of 4×10^9 and zero bytes, now compares calls stored as int8 and two bits each. Every local fit explicitly selects HE; the model grid and convergence settings are held fixed. Method-first order and thread positions are balanced; storage order is balanced across the six pairs. Each fit starts in a fresh process, with tiny compilation/cache warm-ups excluded. Local time includes startup, imports, input, fitting and output. Official time sums its two native stages and excludes subsequent parsing and compression; RSS is their larger stage peak, so the timing boundaries are not perfectly symmetric.

All runs use the 16-GB Apple M2 Pro, AC power and Low Power Mode disabled. Local

BLAS/OpenMP environment limits are one with a Numba pool ceiling of eight; `n_threads` requests one or four workers. The official Mac executable receives the corresponding environment limit and native `--max-threads` argument. The fitted data and drivers are frozen. The reported cross-fitted mixmogam fits were rerun on 7 October with the revision that threads their fold fits' genotype products. The in-sample fits, which that change does not affect, and the storage fits are 2.0.0.dev6 reruns of 6 October. The official fits come from the original 2.0.0.dev3 experiment on the same inputs. Repeated timings on two phenotypes measure computational variability only.

3.5 Historical structure and environmental experiments

The archived structure experiment used 1,300 samples, 20,000 markers, and five chromosomes, with both unstructured genotypes and a four-population generator labelled $F_{ST} = 0.3$ in the archive. We keep that label as a generator setting. Six phenotype replicates shared each genotype panel. Genetic signal was generated outside the last chromosome from a polygenic component of variance share 0.4 and ten large QTLs of variance share 0.1. The last chromosome has no generating marker effects, making it useful for a null-chromosome diagnostic. Sparse QTLs, realized cross-chromosome correlations, and variance estimation mean that this is not the known-Gaussian-covariance experiment above. Loading quintiles use the leading ten kinship eigenvectors.

The archived coalescent experiments used mutation and recombination rates 10^{-8} per base pair per generation, effective population size 10^4, and minor allele frequency above 1% (Kelleher et al., 2016; Baumdicker et al., 2022). Scenario S2 contains two demes separated 400 generations earlier, 400 diploids per deme, and 100 independently simulated blocks of 200 markers. Its environmental exposure is the standardized deme indicator, which is strongly correlated with genotype. The raw trait and standardized exposure are each multiplied by $\sqrt{1/2}$, added, and then standardized together. The original genetic variance target of 0.25 becomes 0.125 before final standardization; the environmental variance target is 0.5. Realized component variances and cross-covariances can change their shares of the final phenotype variance. This is a strong quantitative-trait stress test.

We report S2 statistics in blocks without a designated large QTL. Those blocks still contain infinitesimal effects. We therefore call them *QTL-free blocks*, not strict-null blocks, and do not interpret their detection count as a formal false-positive rate. The same qualification applies to S1 and S3, which have $n = 800$ and $4,000$, $m = 20,000$ and $50,000$, and 20 and 30 large QTLs, respectively. Their target genetic variance share is 0.5, divided equally between a polygenic background and large QTLs. Each scenario has only three independent simulation seeds.

3.6 Summaries, exclusions, and reproducibility

Genomic inflation is summarized as the median chi-square statistic divided by the median of χ^2_1, approximately 0.455. This statistic mixes model error, true signal, and sampling variation. Loading-stratified rejection fractions are more informative about the denominator hypothesis. For the six historical phenotype replicates, Figure 3 uses Student-t intervals for the mean with five degrees of freedom; they describe variation among phenotypes on a fixed genotype panel. S2 displays all three seeds as well as their means.

The archived comparison with the external LDAK program used a faulty PLINK encoding path. A matched-input claim cannot survive that failure. We exclude its empirical results, concordance coefficients, and real-data calibration curves from this revision. The underlying RegMap resource (Horton et al., 2012) remains suitable for a future controlled benchmark. Section 4.13 uses the panel only for a prespecified positive-control check.

New numerical results record the generating script, seeds, software versions, requested BLAS

thread limits, genotype hashes, and package baseline commit. Figure generation records hashes of its actual inputs and outputs. Historical archives keep their original source snapshots and version labels.

The matched phensim, HAPNEST, HE-versus-REML and 20,000-marker results (Sections 4.2, 4.6, 4.7 and 4.8) are reruns with 2.0.0.dev6, except the 20,000-marker cross-fitted fits, rerun once more on 7 October (Section 3.4). Each repeated the mixmogam methods on the archived inputs, after checking the genotype files against their export hashes. The official LDAK-KVIK outputs, logs and process measurements of the original runs were reused unchanged. The reruns ran one after another on 6 October 2026, three days after the official runs, each timed fit waiting for a one-minute load average below 5. Host conditions matter: the original 20,000-marker runs' frozen source, labelled 2.0.0.dev3, equals 2.0.0.dev4 (commit 0fa3da8) apart from its version string, and that identical code took 33.22 seconds on 3 October and 24.1 seconds on 4 October for the same fit. On 6 October exact LOCO and BOLT-inf, whose results did not change, took 1.07–1.16 times as long as in the reruns with commit e4089d8 of 5 October. On one host and day, alternating the versions on one 800-sample case, 2.0.0.dev6 took 1.07 (exact LOCO) and 1.03 (HRATT) times as long as e4089d8 (archive `20261006-same-day-e4089d8-dev6`). Cross-program time ratios therefore include day-to-day and load differences, and Section 4.8 compares dev5 and e4089d8 on one day. The earlier reruns and the original archives remain as provenance.

4 Results

4.1 A nearly correct average can conceal conditional error

Table 2 and Figure 1 show the new conditional experiment. With no frequency differentiation, the scalar approximation is close to the known-covariance test: its analytic rejection probability is 0.99% overall and 1.00% versus 0.97% in the extreme loading quintiles. At $\theta = 0.05$, the overall value remains 1.00% after rounding, but the extreme quintiles separate to 1.19% and 0.73%. The average looks right only because deviations in opposite directions cancel.

At $\theta = 0.20$, the scalar approximation rejects 1.25% overall, with 2.21% in the lowest loading quintile and 0.31% in the highest. The corresponding analytic rejection probabilities at a nominal 5×10^{-8} are 27.4 and 0.025 times nominal. These tail ratios follow from the fully specified Gaussian model and Equation (10), and show why a denominator should be assessed at the threshold that will be used.

The rank-four correction substantially reduces the loading gradient in these panels. At $\theta = 0.05$, its extreme-quintile probabilities are 1.010% and 1.005%; at $\theta = 0.20$, they are 0.998% and 0.992%. Its overall analytic rejection ratios at 5×10^{-8} range from 0.97 to 1.06 across the three panels. Held-out root-mean-square relative denominator-ratio errors are 1.36–1.53%. In these designs the leading structure directions explain most of the approximation error.

The empirical rejection fractions agree with their analytic targets within the script's stated numerical and Monte Carlo tolerance. For the exact denominator, the overall empirical fractions range from 1.006% to 1.007%, with standard errors approximately 0.010 percentage points. There is no fitting uncertainty in this comparison: a score with an exact denominator must have the stipulated marginal law. The simulation checks the implementation of that construction and quantifies the sampling variation of the reported summaries.

Table 2: New known-covariance experiment. Rejection probabilities at a nominal 1% threshold, expressed as percentages. "Simulated" is the overall mean ± one Monte Carlo standard error across 2,000 independent phenotype replicates. q1 and q5 are analytic probabilities in the extreme loading quintiles among 970 held-out audit variants.

		All audit variants		Analytic	
θ	Denominator	Analytic	Simulated	q1	q5
0	Exact	1.00	1.01 ± 0.01	1.00	1.00
0	Scalar	0.99	1.00 ± 0.01	1.00	0.97
0	Spectral, rank 4	0.99	0.99 ± 0.01	0.99	0.99
0.05	Exact	1.00	1.01 ± 0.01	1.00	1.00
0.05	Scalar	1.00	1.01 ± 0.01	1.19	0.73
0.05	Spectral, rank 4	1.01	1.02 ± 0.01	1.01	1.01
0.2	Exact	1.00	1.01 ± 0.01	1.00	1.00
0.2	Scalar	1.25	1.26 ± 0.01	2.21	0.31
0.2	Spectral, rank 4	1.00	1.00 ± 0.01	1.00	0.99

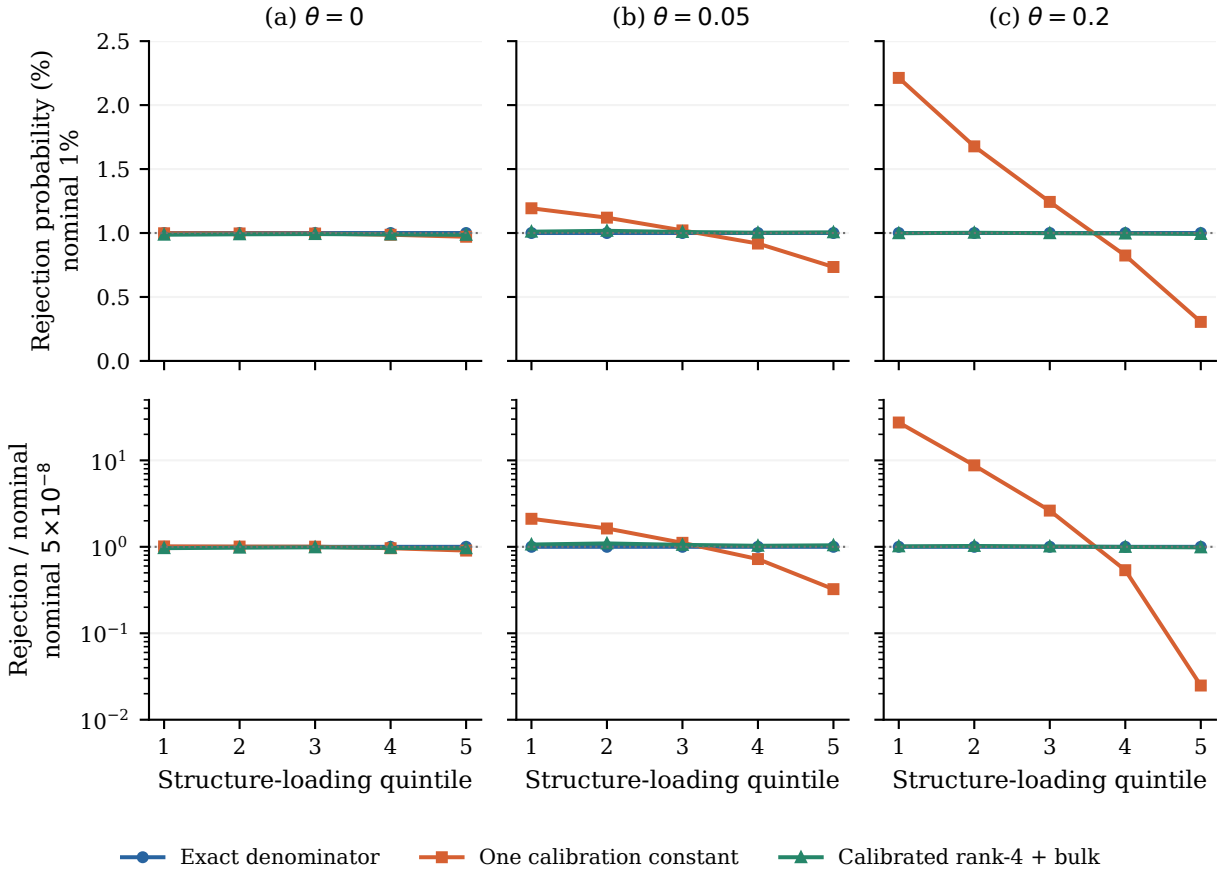


Figure 1: Conditional calibration under known covariance. The top row shows analytic rejection probabilities at a nominal 1%; the bottom row shows the ratio of analytic to nominal rejection probability at 5×10^{-8} on a log scale. Each column is one fixed genotype panel. Candidate variants are held out from approximation calibration and ordered by loading on the leading three background eigenvectors. The spectral method uses four exact eigenvectors and a flat bulk. Columns share vertical scales.

4.2 Matched reference comparisons depend on mean adjustment

All 384 method runs returned valid p-values for every retained variant: 288 mixmogam runs repeated with 2.0.0.dev6 and the 96 original official runs. mixmogam HRATT reported incomplete cross-validation in six unadjusted confounded mixed-trait runs, one of which also had incomplete LOCO fits (two in the original runs). These remain in the summaries. In the original archive, an independently regenerated structured/PC-adjusted case reproduced the serialized inputs and all four methods' p-values exactly using frozen source snapshots. Exact LOCO and BOLT-inf reproduce every rejection rate of the earlier reruns with commit e4089d8: BOLT-inf's p-values are identical, and exact LOCO's moved by at most 1.3×10^{-6} in $\log_{10} p$, from the refined REML search of 2.0.0.dev6. HRATT's cross-fitted scores (Section 2.13) changed 0.26% of its decisions at 1%, its rejection rates by at most 0.10 percentage points and its detections by at most three of a cell's 60 causal markers.

Figure 2 separates genetic-null rejection, mixed-trait rejection on chromosome 6, and causal-marker detection. In unstructured global-null cells, mean rejection at 1% is approximately 1.00–1.06% across methods and LD settings. Under structure and the mixed trait, LDAK-KVIK's null-chromosome rejection is 1.87% without local LD and 1.82% with LD.

Omitting the environmental mean shift produces the largest disagreement. With LD, LDAK-KVIK gives $\lambda_{\mathrm{GC}} = 3.455$ and 15.83% genetic-null rejection at 1%; exact LOCO and mixmogam BOLT-inf give 1.34% and 1.25%, and mixmogam HRATT gives 0.64% (Table 3). LDAK-KVIK rejection is 21.5–25.2% in four panels and 1.52–1.82% in two. Some inflated runs reach LDAK-KVIK's lower heritability boundary after revision of the initial estimate; we have not identified the cause. The mixmogam deflation also fails nominal calibration.

Adding the same two PCs reduces LD global-null rejection to 0.92% (mixmogam HRATT) and 0.97% (LDAK-KVIK). Mixed-trait null-chromosome rejection is 1.17% and 1.05%. They detect 29 and 30 of the 60 causal markers; ten causal markers per panel cannot resolve a power difference. In these panels, explicit mean adjustment changes the comparison more than the choice of method does.

4.3 Historical scans show a related pattern, with weaker identification

The archived strong-structure scan exhibits the same qualitative gradient (Table 4, Figure 3). For the scalar infinitesimal implementation, overall λ_{GC} is 0.97, but the lowest and highest loading quintiles have $\lambda_{\mathrm{GC}} = 1.34$ and 0.64, and rejection fractions of 2.45% and 0.11% at 1%. The historical spectral correction gives 1.14% and 1.06% in those quintiles, close to the dense LOCO results of 1.12% and 1.11%. The historical HRATT implementation retains a pronounced gradient.

These results support the mechanism but do not prove it a second time: variance fitting, sparse generating effects, calibration-marker selection and the earlier implementation all contribute to the final statistics. The new experiment identifies denominator error because it removes those features, and for the same reason says less about the complete scan.

4.4 Covariance adjustment and environmental adjustment are different

In the historical S2 stress test, the leading PC predicts deme membership with squared correlation approximately 0.97. Without kinship adjustment, QTL-free blocks have $\lambda_{\mathrm{GC}} = 11.90$; the mean statistic among variants with the highest environmental loading is 63.24. Dense LOCO reduces these summaries to 1.24 and 2.27, but substantial residual structure remains (Table 5, Figure 4). Including PC1 reduces them further to 1.09 and 1.13. The corresponding all-variant λ_{GC} falls from 1.27 to 1.11.

Table 3: LD panels in the matched comparison. Rates are percentages; ± is one Monte Carlo standard error across six panels. Inflation uses the global genetic null. Rejection columns use that null and the mixed trait's null chromosome. Detection uses $0.05/m$.

Method	λ_{GC}	Global null	Null chromosome	Detection
Unstructured				
Exact LOCO	1.012	1.03 ± 0.07	0.83 ± 0.15	46.67 ± 3.33
mixmogam BOLT-inf	1.016	1.03 ± 0.09	0.88 ± 0.14	45.00 ± 3.42
mixmogam HRATT	1.015	1.01 ± 0.08	0.98 ± 0.14	48.33 ± 3.07
LDAK-KVIK	1.018	1.06 ± 0.08	1.05 ± 0.14	50.00 ± 3.65
Structured				
Exact LOCO	0.987	0.91 ± 0.07	1.00 ± 0.20	41.67 ± 3.07
mixmogam BOLT-inf	1.006	0.92 ± 0.09	1.20 ± 0.21	41.67 ± 4.01
mixmogam HRATT	1.004	0.93 ± 0.09	1.10 ± 0.11	50.00 ± 3.65
LDAK-KVIK	0.983	0.93 ± 0.06	1.82 ± 0.77	48.33 ± 3.07
Structured + environment				
Exact LOCO	1.115	1.34 ± 0.07	1.08 ± 0.20	38.33 ± 4.01
mixmogam BOLT-inf	1.114	1.25 ± 0.08	1.32 ± 0.20	38.33 ± 4.01
mixmogam HRATT	0.947	0.64 ± 0.07	0.72 ± 0.22	30.00 ± 4.47
LDAK-KVIK	3.455	15.83 ± 4.51	6.27 ± 4.77	40.00 ± 3.65
Same trait + 2 PCs				
Exact LOCO	1.009	0.94 ± 0.07	1.00 ± 0.19	40.00 ± 3.65
mixmogam BOLT-inf	1.011	0.94 ± 0.07	1.15 ± 0.23	40.00 ± 3.65
mixmogam HRATT	1.010	0.92 ± 0.07	1.17 ± 0.07	48.33 ± 4.01
LDAK-KVIK	0.940	0.97 ± 0.06	1.05 ± 0.16	50.00 ± 3.65

Table 4: Historical simulated strong-structure panel, six phenotype replicates. q1 and q5 denote loading quintiles. Rejection fractions are for the chromosome without generating effects, under the qualifications in Section 3.5. Archived implementation results.

		λ_{GC}		Rejection (%)	
Method	λ_{GC}, all	q1	q5	q1	q5
Exact LOCO	1.00	1.05	1.02	1.12	1.11
Infinitesimal, constant	0.97	1.34	0.64	2.45	0.11
Infinitesimal, spectral	1.00	1.05	1.01	1.14	1.06
HRATT	1.02	1.37	0.73	2.92	0.11

Table 5: Historical S2 environmental stress test, means over three seeds. Loading refers to squared correlation with the deme exposure. Detection counts are significant QTL-free blocks, not false positives under a strict null; these blocks also contain polygenic background effects. Power is the fraction of designated QTL-containing loci detected under the archived Bonferroni rule.

		Mean χ^2		QTL-free	Locus
Method	λ_{GC}, QTL-free	Bottom half	Top 1%	detections	power
Linear model	11.90	2.20	63.24	90.33	1.00
LMM, no LOCO	0.96	0.87	1.77	0.00	0.13
Exact LOCO	1.24	1.13	2.27	0.67	0.20
Exact LOCO + PC1	1.09	1.10	1.13	0.67	0.20

The non-LOCO scan has all-variant $\lambda_{GC} = 0.98$ and QTL-free $\lambda_{GC} = 0.96$, yet its upper-loading mean remains 1.77 and its locus power is lower than LOCO's (0.13 versus 0.20). Apparent aggregate calibration is therefore compatible with both residual environmental structure and signal attenuation. Separating the two needs a factorial simulation that varies the mean and covariance mechanisms independently (Section 6).

A related interpretation problem appears in S1 and S3. Replacing fixed 200-marker intervals with LD-aware boundaries reduces the number of QTL-free detections. In the recorded paired checks, dense LOCO's mean count changes from 1.33 to zero across three S1 seeds; the single paired S3 check changes from 17 to 11, consistent with large-QTL signal crossing arbitrary boundaries. With a polygenic background the remaining detections need not be errors, and other block definitions change the unit of discovery. Methods for constructing approximately independent LD blocks are useful here (Berisa and Pickrell, 2016; Privé, 2022), but evaluation must still define which biological or statistical null a "false locus" is meant to represent.

4.5 Correctness and computational evidence

The initial correctness review recorded 117 passing local tests. Its installed wheel check on Python 3.10 with NumPy 1.26.4 and SciPy 1.11.4 passed 71 tests, with five skips for unavailable optional HDF5/plotting components. These checks include independent numerical references and reproductions of specific defects. The corrected contracts cover PLINK coding, sample alignment and allele counts, BLUP scaling, variance fitting, weighted kinships, window complements, permutation geometry, and effect/standard-error export.

One allocation change has a directly recorded before/after comparison. For 800 samples, 1,000 markers, ten LOCO groups, and 80 right-hand sides, five warm matrix-product repetitions gave identical output arrays. The measured peak allocation fell from 7,235,824 to 2,625,120 bytes, a 64% reduction. Python allocation tracing excludes the pre-existing genotype cache and native BLAS allocations, so this is not whole-process peak memory.

Older timing archives show why batching and streaming help. One internal comparison recorded 0.27 seconds for a batched EMMAX scan versus 33.7 seconds for a per-variant least-squares reference at $n = 2,000$, $m = 10,000$ (about 123-fold). Historical S3 times were 221 seconds for 25-group dense LOCO and 18 seconds for the infinitesimal two-step path; a 2.7-second non-LOCO scan excluded its preceding fit. These workloads differ in both statistical procedure and accounted work, and were measured on a development laptop under concurrent activity.

At $n = 10,000$, a historical stochastic-Lanczos fit obtained $\delta = 0.91724$ versus 0.91639 for a full fit, a 0.09% difference in one case. A historical scan based on a truncated spectrum still produced $\lambda_{GC} = 1.30$, versus 0.85 for the full non-LOCO scan. Accurate likelihood approximation, accurate association denominators, and calibrated tests are separate requirements.

4.6 Measured time, resident memory, and simulation scale

New timings use an Apple M2 Pro with 16 GB RAM, Python 3.14.6, NumPy 2.4.6 and Numba 0.66.0, with one thread per method. mixmogam HRATT uses its compiled kernels. Across the 96 primary inputs its median time is 2.27 seconds versus 0.74 seconds for LDAK-KVIK, with median RSS 260.8 versus 40.1 MiB (Table 6); the original 2.0.0.dev2 runs took 2.57 seconds and 319.5 MiB, and the reruns with commit e4089d8 1.89 seconds and 199.1 MiB on a faster day. Startup and loading matter at this size, the two programs were timed on different days, and the native Mac LDAK build lacks MKL.

Figure 5 and Table 7 extend the controlled LD workloads to 4,000 samples and 24,000 markers.

Dense exact scans become substantially more expensive, although at 4,000 samples they now take about 31 seconds instead of 91. LDAK-KVIK retains lower observed time and RSS than mixmogam HRATT. Because n and m both increase, the slopes do not identify a sample-size complexity exponent.

Table 6: Primary end-to-end resources at n=800 and m=6,000. Each row covers the same 96 inputs. mixmogam rows are 2.0.0.dev6 reruns; LDAK-KVIK rows are the original measurements. LDAK-KVIK time sums both stages; its RSS is the larger stage peak. Common data preparation is excluded.

Method	Runs	Median s	Range s	Median MiB	Max MiB
Exact LOCO	96	1.72	1.32–4.64	268.6	299.1
mixmogam BOLT-inf	96	1.78	1.43–3.33	242.0	264.1
mixmogam HRATT	96	2.27	1.52–7.66	260.8	281.1
LDAK-KVIK	96	0.74	0.25–1.93	40.1	45.6

Table 7: Median seconds for the workload extension and maximum peak RSS at the largest workload. Mixed traits have latent LD correlation 0.8. Column labels show samples/markers; preparation is excluded.

Method	800/6,000	2,000/12,000	4,000/24,000	Max MiB
Unstructured				
Exact LOCO	1.57	3.21	30.53	992
mixmogam BOLT-inf	1.73	2.33	6.27	447
mixmogam HRATT	2.19	3.65	8.28	530
LDAK-KVIK	0.72	1.39	3.44	154
Structure + environment + 2 PCs				
Exact LOCO	1.86	3.09	31.14	854
mixmogam BOLT-inf	1.90	2.50	5.34	436
mixmogam HRATT	2.35	3.28	8.35	544
LDAK-KVIK	0.84	1.46	5.33	153

The HAPNEST-model kernel supplies larger cohorts without generating a full ancestral graph. At 2,000 samples and 12,000 variants, median warm generation time falls from 22.28 seconds for NumPy to 0.48 seconds for Numba, a 46.8-fold ratio; all paired output hashes agree. Streaming 100,000 samples takes 25.73–28.40 seconds and at most 198.6 MiB process RSS across three repeats (Table 8). These times include checksums but exclude loading the small reference and first-call setup; we did not run the upstream Julia program. At 50,000 samples a full int8 matrix requires 600 million bytes; RAM and memory-mapped modes retain much more resident memory than streaming.

At n=10,000 and m=12,000, tiling the phenotype factor reduces median process peak RSS from 1,607.5 to 320.2 MiB across three matched seeds; median time changes from 1.20 to 0.89 seconds. All returned components agree with the dense formula within 2×10^{-12}. Tiled PLINK export reduces median peak RSS from 811.2 to 308.3 MiB and yields identical BED hashes. Its median time is 0.81 versus 0.88 seconds, with substantial variation across repetitions; the clearest benefit is memory. The dense encoding baseline omits small metadata-writing costs, whereas the tiled writer includes them. These preparation measurements include the resident genotype and interpreter.

Table 9 holds m=12,000 fixed and extends the matched association comparison to n=50,000, with mixmogam HRATT at its default REML. At 50,000 samples the 2.0.0.dev6 reruns took 45.27 and 48.38 seconds, against 38.19 and 40.20 with commit e4089d8 on a faster day, 137.59 and 318.02

seconds in the original 2.0.0.dev2 runs and 24.45 and 35.42 seconds for official LDAK-KVIK. Peak RSS fell from 2.93 and 3.16 GiB in the original runs to 1.86 and 1.83 GiB. Fitted heritabilities are unchanged; with cross-fitted scores, inflation moved by at most 0.07 and rejection by at most 0.2 percentage points. These are single-panel observations (Table 1). All four rerun mixmogam HRATT fits at these larger sizes report completed cross-validation and LOCO convergence. Both programs return valid p-values for all markers.

Table 8: HAPNEST-model generation at 12,000 variants: three independent seeds, one thread, warm kernels, with output checksums. Peak RSS includes interpreter and loaded reference. Modes use identical draws for a fixed seed; memory-mapped pages can become resident.

Samples	Backend / output	Median s	Range s	Max MiB
2,000	numba / batches	0.48	0.43–0.81	164.6
2,000	numpy / batches	22.28	21.08–22.28	121.1
10,000	numba / batches	2.31	2.21–2.58	173.5
50,000	numba / array	13.62	12.93–13.94	757.3
50,000	numba / batches	13.32	11.88–13.35	183.0
50,000	numba / memmap	14.24	13.40–14.46	756.6
100,000	numba / batches	26.13	25.73–28.40	198.6

Table 9: HAPNEST-model workloads, m=12,000, one realization per cell. mixmogam HRATT (2.0.0.dev6 rerun) uses REML; LDAK-KVIK rows are the original measurements on the same inputs. Time is end-to-end seconds; RSS is peak MiB. Inflation and 1% rejection use the chromosome with no generating genetic effects. Structured cells include an environmental effect and two genotype PCs. These descriptive values have no replicate-based uncertainty estimate.

Samples	Cell	Program	Seconds	MiB	λ_{GC}	Rejection (%)
10,000	Unstructured	mixmogam	9.97	552	0.976	0.85
10,000	Unstructured	LDAK	4.20	220	0.931	0.85
10,000	Structure + PCs	mixmogam	9.77	656	1.131	1.00
10,000	Structure + PCs	LDAK	4.06	221	0.992	1.10
50,000	Unstructured	mixmogam	45.27	1908	0.949	0.95
50,000	Unstructured	LDAK	24.45	627	0.963	1.10
50,000	Structure + PCs	mixmogam	48.38	1877	1.047	1.10
50,000	Structure + PCs	LDAK	35.42	653	0.965	1.30

4.7 Estimator choice and implementation improvements have different effects

The implementation-only comparison, between two 2.0.0.dev3 sources on 3 October (archive `20261003-kvik-efficiency`), held REML, model selection and convergence settings fixed. At 50,000 samples and 12,000 markers, three paired repetitions reduced median complete-fit time from 105.21 to 90.33 seconds without structure, and from 106.29 to 90.31 seconds with structure, environment and PCs: reductions of 14.1% and 15.0%. Across all 12 paired comparisons, including the 10,000-sample panels, saved variational coefficients and fit diagnostics were exact; the maximum absolute p-value difference was 1.45×10^{-15}. Removing redundant products and allocations helped, but repeated kinship products for variance fitting remained a major cost.

In a separate one-thread comparison on the two 50,000-sample, 12,000-marker panels, rerun with 2.0.0.dev6, choosing HE reduced median complete-fit time from 43.29 to 21.21 seconds and

from 48.79 to 24.83 seconds, respectively. These are within-experiment comparisons; the REML times should not be substituted for the baseline of the preceding experiment. The original runs on 3 October, with older code, took 76.13 and 26.28 seconds, and 73.57 and 27.20; Section 4.9 separates the code effect from the host. HE changed the fitted covariance ratio from 0.49543 to 0.51260 and from 0.50125 to 0.52087, so the time saving comes with a different estimate. REML also needs more memory: 1.68 and 1.54 against 1.08 and 1.02 GiB peak RSS.

The independent known-covariance study supports the narrower claim that the projected moment equations target their stated coefficients. Expected unconstrained moments recover v_g and v_e within 1.2×10^{-9} and 1.2×10^{-10} over 12 analytic checks. Across phenotype draws, HE mean ratios at targets 0.3 and 0.6 are 0.3026 and 0.5930 without structure, and 0.2972 and 0.5968 with PCs; corresponding RMSE values range from 0.0508 to 0.0698. At the zero boundary, means are positive (0.0148 and 0.0165), as for the constrained dense comparator, a finite-sample boundary bias. No estimator failures occurred. In the separate global-null scans, both HE and REML reject 0.938% and 0.948% of markers at the 1% threshold in the two cells, respectively, with all 64 fits converged, on four fixed genotype panels.

4.8 Twenty thousand markers: speed, memory and numerical stability

All mixmogam fits and the 12 reused official fits are complete, and every local cross-validation and LOCO fit converged. The mixmogam fits use the default cross-fitted LOCO scores, repeated with in-sample scores (`loco_folds=1`), which reproduce the summaries of the reruns with commit e4089d8 exactly. With cross-fitted scores h^2 is unchanged, and the null chromosome's λ_{GC} moved from 0.951 to 0.946 and from 0.978 to 0.984 (Table 11).

Four requested workers sped the cross-fitted fits up 1.40 and 1.48 times on the unstructured and PC-adjusted confounded panels, to 33.15 and 29.21 seconds, and the in-sample fits 1.57 and 1.47 times, to 21.74 and 20.70 seconds (Table 10). In 2.0.0.dev6 the variational sweep threaded its genotype products only from six model columns on, and the cross-fit fits five, one per fold: four workers then gained only 1.27 and 1.33-fold, to 34.63 and 31.64 seconds, at a CPU-to-wall ratio of 1.2 against 1.4 with in-sample scores. Six was the smallest shape probed when that threshold was set, not a break-even: at 50,000 samples, threaded products sped fitted sweeps up 1.31–1.40-fold for two to eight columns alike, so the column threshold was dropped. In same-day pairs, four-worker cross-fitted fits took 0.90–0.94 times as long as with 2.0.0.dev6. On one thread, whose code and results are unchanged, cross-fitting cost 1.28 and 1.38 times the in-sample time on 6 October; on four, 1.52 and 1.41 times across the two days. Host conditions moved the one-thread cross-fitted fits by 1.05 and 1.02 between those days, and the in-sample fits by 1.04–1.09 between 5 and 6 October. The official executable took 25.45 and 30.09 seconds on 3 October, when the host was slower (Section 3.6), so these times cannot be compared directly. A same-day check on 5 October ran the first panel's four-worker fit with both versions: 24.6 seconds and 4.4–5.1 GiB peak RSS with dev5's cache, 20.5 seconds and 1.44 GiB with e4089d8. On a 12,000-marker panel, one-thread REML took 31.8 and 38.2 seconds; exact LOCO, unchanged, took the same time and memory under both versions, with identical p-values. mixmogam's median peak RSS of 1.4 GiB is about 2.2 times the official executable's 0.65 GiB. Optional parallelism helps, now also for the cross-fitted default, but mixmogam still uses more memory.

The numerical audit preserves its original elementwise tolerances, $\mathrm{rtol} = 10^{-6}$ and $\mathrm{atol} = 10^{-8}$. All 24 repeated-run comparisons within a method and worker count are exact: eight cross-fitted, eight in-sample and eight official. Selected exponents and priors, integer iteration counts and convergence flags agree across counts. With cross-fitted scores, one and four workers differ by about the in-sample amounts below: on the PC-adjusted panel by at most 1.2846×10^{-7} in effects,

Table 10: Complete fits at 50,000 samples and 20,000 markers on identical inputs: mixmogam with its default cross-fitted LOCO scores (the revision after 2.0.0.dev6, 7 October) and with in-sample scores (2.0.0.dev6, 6 October; unaffected by the later change), and the original official fits (3 October). Wall time and peak RSS are median [minimum, maximum] of three fresh-process repetitions on each fixed panel. Speedup is relative to the same method's one-thread median. Timing boundaries are defined in Section 3.4; ratios across rows span days with different host conditions. The two largest four-worker cross-fitted times (47.49 and 39.23 seconds) coincided with high system time, which points to memory compression.

Method	Threads	Wall seconds	Speedup	Peak RSS (GiB)
Unstructured				
mixmogam HE	1	46.25 [44.73, 46.40]	1.00	1.379 [1.346, 1.396]
mixmogam HE	4	33.15 [29.72, 47.49]	1.40	1.355 [1.314, 1.387]
mixmogam HE, in-sample	1	34.18 [34.16, 34.80]	1.00	1.383 [1.382, 1.445]
mixmogam HE, in-sample	4	21.74 [21.72, 21.91]	1.57	1.428 [1.404, 1.440]
LDAK-KVIK	1	33.49 [33.43, 34.90]	1.00	0.651 [0.651, 0.652]
LDAK-KVIK	4	25.45 [24.51, 28.16]	1.32	0.652 [0.651, 0.652]
Structure, environment, two PCs				
mixmogam HE	1	43.09 [40.68, 46.61]	1.00	1.427 [1.425, 1.476]
mixmogam HE	4	29.21 [28.71, 39.23]	1.48	1.399 [1.294, 1.491]
mixmogam HE, in-sample	1	30.52 [30.32, 31.17]	1.00	1.418 [1.380, 1.451]
mixmogam HE, in-sample	4	20.70 [20.08, 20.86]	1.47	1.459 [1.435, 1.463]
LDAK-KVIK	1	39.63 [36.71, 40.54]	1.00	0.654 [0.652, 0.656]
LDAK-KVIK	4	30.09 [28.25, 32.61]	1.32	0.653 [0.651, 0.654]

3.0606×10^{-10} in standard errors, 1.3189×10^{-5} in p-values and 6.45×10^{-5} in log-base-ten p-values. One-thread results equal those of 2.0.0.dev6 bit for bit. In 2.0.0.dev6, whose fold fits' products were not threaded, one and four workers gave identical association arrays; only the cross-validation fit's coefficients, which select the prior, differed beyond the strict tolerance. With in-sample scores the differences equal those of the reruns with commit e4089d8. On the unstructured panel they are of the original run's size. On the PC-adjusted panel they are about tenfold: the largest are 1.3446×10^{-7} for association effects, 2.1913×10^{-10} for standard errors and 1.9618×10^{-5} for p-values, and 1.06×10^{-4} in log-base-ten p-values, against 9.7892×10^{-9}, 3.0826×10^{-11}, 1.4312×10^{-6} and 1.15×10^{-5} in the original run. None of the six official comparisons exceeds the tolerance. Covariates are now removed by subtracting rank-q corrections from products of unprojected genotypes (Equation (14)), which probably amplifies summation-order differences when PCs explain genotype variance. All-variant significance masks agree at 0.05, 0.01, 0.001, 0.05/20,000 and 5×10^{-8} in all 60 checks, 30 cross-fitted and 30 in-sample. The four-versus-one tolerance failures, six per set of fits, still count as failures, and other data could cross a threshold.

Table 11 reports descriptive chromosome-6 diagnostics from four-worker cross-fitted fits. Local HE ratios are approximately 0.52 and 0.50, whereas the official values are approximately 0.25. The different partitioning, large-effect handling and fitted covariance definitions preclude interpreting this discrepancy as bias against a shared target. Both programs detect all ten designated QTLs at both tested stringent thresholds, a saturated result with little power to distinguish methods.

The separate storage experiment replaces the cache experiment, since no cache remains (Table 12). With dev5, disabling the cache had cut peak RSS by 63% at 13% and 25% longer wall time. Calls stored two bits each about halve the remaining peak, to 0.77 and 0.79 GiB (0.73 GiB with in-sample scores on 5 October) against the official executable's 0.65 GiB, at the same wall time. All six storage pairs have exactly equal saved association arrays, variational coefficients and

Table 11: Descriptive diagnostics on the 3,300-marker genetic-null chromosome of each 20,000-marker panel, using four requested workers. Rejections are percentages. Each entry is unchanged across three timing repetitions of one phenotype. Reported h^2 follows each program's own covariance-fitting convention and is not a common estimand.

Method	Reported h^2	λ_{GC}	$p < 0.05$	$p < 0.01$	$p < 0.001$
Unstructured					
mixmogam HE	0.5224	0.9457	4.848	0.818	0.091
LDAK-KVIK	0.2461	0.9435	4.788	0.879	0.061
Structure, environment, two PCs					
mixmogam HE	0.5049	0.9837	5.727	1.273	0.273
LDAK-KVIK	0.2479	0.9189	4.788	1.333	0.303

nested fit diagnostics. The calls, prepared per-marker values, Gram matrices and fitting workspaces stay in memory, so the analysis is not out-of-core.

Table 12: Separate storage experiment at 50,000 samples and 20,000 markers, explicit HE, cross-fitted scores and four workers, 2.0.0.dev6. Entries are median [minimum, maximum] over three matched repetitions. The only change is how the calls are stored: one byte or two bits each.

Calls	Wall seconds	Peak RSS (GiB)
Unstructured		
int8 calls	29.80 [29.24, 29.96]	1.449 [1.405, 1.468]
Two-bit calls	29.86 [28.64, 33.07]	0.771 [0.767, 0.814]
Structure, environment, two PCs		
int8 calls	28.61 [28.57, 28.82]	1.467 [1.411, 1.473]
Two-bit calls	27.36 [27.14, 27.60]	0.785 [0.769, 0.801]

Resource interpretation has two additional limits. In the original run, system-wide VM observations spanning the period after primary run nine through the end of all fits record 32,093 additional swap-in pages and 111,568 swap-out pages. They cannot identify which program or fit caused swapping. The dev5 reruns recorded 7,682 swap-in pages and no swap-outs; the later reruns kept no VM counters. Second, the pinned official Mac binary links Accelerate and inspection finds no OpenMP library linkage or matching symbols. Source-level OpenMP directives cannot be assumed active in this executable. These measurements do not establish relative scaling against LDAK's Linux OpenMP/MKL build.

4.9 Work reduction: paired fits and Lanczos accuracy

Table 13 compares frozen 2.0.0.dev4 and 2.0.0.dev5 sources on five seeded synthetic workloads, one per association path. Each source first warmed its own Numba cache on a small panel. Measured fits then ran in fresh processes, two per source, alternating which source ran first. An unrelated session's jobs kept the host at a load average of about 2.4 throughout, so the timings are paired under shared load. Both repetitions of every fit returned identical arrays.

The exact LOCO scan took 13.3 instead of 79.6 seconds. Its CPU time fell less, from 89 to 60 seconds, because the factorizations use several cores where the eigensolver used about one. The first group took 31 factorizations and the other 21 took 11 to 18 (median 13). Variance ratios agreed to a relative 6.1×10^{-7}, and $\log_{10} p$ to 3×10^{-5}; the remaining difference is float32 rounding under a

Table 13: Paired fits of frozen 2.0.0.dev4 and 2.0.0.dev5 sources on simulated panels (samples/markers). Fit seconds are median [minimum, maximum] over two fresh-process repetitions; the ratio compares medians. Peak RSS is the median whole-process value in GiB. Agreement is the largest absolute difference in $\log_{10} p$ between versions. MLMM's default forward-scan precision changed from float64 to float32.

Workload	dev4 s	dev5 s	Ratio	dev4 GiB	dev5 GiB	Agreement
Exact LOCO, 3,000/44,000	79.6 [78.7, 80.5]	13.3 [13.3, 13.3]	6.0	1.14	0.80	3×10^{-5}
MLMM, ten steps, 2,000/50,000	27.4 [27.3, 27.4]	5.5 [5.4, 5.5]	5.0	1.41	1.45	same cofactors
BOLT-LMM-inf, 10,000/30,000	12.9 [12.9, 13.0]	7.3 [7.3, 7.3]	1.8	1.85	1.84	7×10^{-2}
HRATT with REML, 10,000/30,000	14.4 [14.4, 14.4]	9.8 [9.8, 9.8]	1.5	1.91	1.93	5×10^{-2}
HRATT-HE, no cache, four threads, 20,000/20,000	7.7 [7.5, 8.0]	7.3 [7.0, 7.6]	1.1	1.14	0.93	identical

different whitening. Peak RSS fell from 1.14 to 0.80 GiB. MLMM ran five times faster and selected the same cofactors under all three criteria; its new float32 default accounts for part of that gain.

The two-step paths gained less, 1.8-fold for BOLT-LMM-inf and 1.5-fold for HRATT with REML, because their genotype passes were already batched. Their heritability estimate moved from 0.4440 to 0.4369 with the new Lanczos probes. With new calibration markers, $\log_{10} p$ moved by up to 0.07 at $p \approx 4 \times 10^{-31}$. These are Monte Carlo changes, not errors of either version. Uncached HRATT-HE draws neither probes nor calibration markers here, and its p-values were identical. Its peak RSS fell from 1.14 to 0.93 GiB at similar speed. A same-day check at 50,000 samples gives the same picture. With 12,000 markers, REML fits took 31.7 seconds with 2.0.0.dev5 against 49.8 with 2.0.0.dev4. Cached four-worker HE fits with 20,000 markers took 24.0 and 24.2 seconds. Peak RSS was 5.1 GiB for HE in both versions, and 4.1 against 4.2 GiB for REML.

Table 14 supports the new Lanczos defaults. Each panel compares two-step REML with exact REML on a dense kinship formed from the same projected genotype blocks, over independent probe seeds. The table comes from a rerun with commit e4089d8, which reproduced the dev5 study to within 5×10^{-7} in every summary. Twenty-four steps reproduced 96 to within 1.1×10^{-7} in h^2 on every panel. Sixteen steps missed by 1.1×10^{-4} when markers were fewer than samples. The remaining error came from the probes. With six populations at $n = 500$, 12 probes gave a standard deviation of 0.116 over seeds and one error of 0.38; 48 probes reduced these to 0.018 and 0.028. On the other panels the gain was smaller. These six panels, each with one phenotype, test the trace estimator, not the sampling distribution of h^2.

Table 14: Two-step REML heritability at 24 Lanczos steps against exact REML on the same projected genotypes (samples/markers), commit e4089d8. Entries are mean ± standard deviation over six probe seeds (twelve at $n = 500$), with the largest absolute error in parentheses. The last column is the largest change from 24 to 96 steps. In the first simulated $h^2 = 0.95$ panel both solvers stop at the upper limit of the variance-ratio grid.

Panel	Exact h^2	12 probes	48 probes	24 vs 96 steps
Two populations, covariate, 5,000/29,998	0.2435	0.2514 ± 0.0111 (0.0156)	0.2461 ± 0.0080 (0.0165)	1×10^{-14}
Four populations, $F_{ST} = 0.1$, 3,000/19,970	0.5775	0.5697 ± 0.0076 (0.0151)	0.5755 ± 0.0068 (0.0101)	7×10^{-9}
The same with covariates, 3,000/19,971	0.4983	0.4951 ± 0.0247 (0.0449)	0.4979 ± 0.0183 (0.0306)	9×10^{-15}
Simulated $h^2 = 0.95$, 3,000/20,000	1.0000	1.0000 ± 0.0000 (0.0000)	1.0000 ± 0.0000 (0.0000)	0
Simulated $h^2 = 0.95$, 3,000/1,500	0.9505	0.9506 ± 0.0007 (0.0009)	0.9506 ± 0.0002 (0.0004)	9×10^{-8}
Six populations, $F_{ST} = 0.35$, 500/3,910	0.6178	0.6546 ± 0.1165 (0.3822)	0.6210 ± 0.0183 (0.0278)	7×10^{-8}

4.10 Genotype streaming: paired fits and memory growth

Table 15 compares frozen 2.0.0.dev5 with commit d51b4c3, which is e4089d8 with HRATT renamed (Section 2.11), on every association path. Each source runs with its defaults: dev5 caches standardized genotypes wherever they fit its 4×10^9-byte budget, as on every panel here, and d51b4c3 decodes int8 or two-bit calls on every pass. Each fit ran three times per source in fresh processes, the source order rotating, and every measurement waited until the host's one-minute load average was below 5.

The two-step paths need 2.4 to 2.8 times less memory with int8 calls and 3.1 to 4.2 times less with two-bit calls. HRATT-HE ran 4% faster than dev5 on one thread and 14% faster on four. BOLT-LMM-inf, BOLT-LMM and HRATT with REML took 1.19 to 1.23 times as long, since every pass now decodes the calls, and 1.14 to 1.18 times with two-bit calls. Exact LOCO and MLMM, whose code did not change, took the same time. Every repetition returned identical arrays, two-bit and int8 calls gave identical arrays in every workload, exact LOCO and MLMM matched dev5 exactly, and the two-step paths moved by at most 4.2×10^{-5} in $\log_{10} p$. Table 16 varies the number of markers for HRATT-HE at 10,000 samples. Peak RSS grows by 0.36 bytes per genotype with two-bit calls and 1.14 with int8 calls, against 4.06 with dev5's cache and 1.26 without it. d51b4c3 runs slightly faster than dev5 with its cache, 16.3 against 17.9 seconds at 80,000 markers, and needs 40% less time than dev5 without it.

The steps between the two versions are archived separately: `20261004-genotype-streaming` (12aa504, covariates no longer projected per marker), `20261004-gram-cache` (Gram matrices stored in float32 and cached in part, which made cross-validation 1.5 and 2.5 times faster for BOLT-LMM and HRATT where float64 Grams had exceeded the budget) and `20261004-packed-calls` (two-bit calls, no float cache).

Table 15: Paired fits of frozen 2.0.0.dev5 and commit d51b4c3 on simulated panels (samples/markers), each with its defaults: dev5 with its float genotype cache, d51b4c3 with int8 calls or, on the two-step paths, two-bit calls. Fit seconds and whole-process peak RSS (GiB) are medians over three fresh-process repetitions per source, each started below a one-minute load average of 5; repetitions were within 8% of their median, 14% for four-thread fits. Agreement is the largest absolute difference in $\log_{10} p$ between versions; two-bit and int8 results were identical.

Workload	dev5 s	int8 s	Two-bit s	dev5 GiB	int8 GiB	Two-bit GiB	Agreement
Exact LOCO, 3,000/44,000	15.3	14.9	–	0.83	0.84	–	identical
MLMM, ten steps, 2,000/50,000	6.1	6.2	–	1.44	1.45	–	same cofactors
BOLT-LMM-inf, 10,000/30,000	8.2	10.1	9.6	1.88	0.79	0.60	8×10^{-6}
BOLT-LMM, 10,000/30,000	19.0	22.8	21.6	3.11	1.16	0.95	4×10^{-5}
HRATT with REML, 10,000/30,000	10.6	12.7	12.3	1.95	0.82	0.62	1×10^{-5}
HRATT-HE, 10,000/30,000	6.1	5.8	5.8	1.77	0.65	0.43	9×10^{-6}
HRATT-HE, four threads, 20,000/20,000	6.3	5.4	5.6	2.24	0.81	0.54	2×10^{-5}

Table 16: HRATT-HE at 10,000 samples: median fit seconds / peak RSS (GiB) over three repetitions, for dev5 with its default float genotype cache, which held every panel, and with the cache disabled, and for the current source with int8 and two-bit calls. The last row is the least-squares growth of peak RSS per genotype. Agreement compares dev5's default with the current source; two-bit and int8 results were identical.

Markers	dev5 cache	dev5 no cache	int8	Two-bit	Agreement
10,000	2.0 / 0.81	3.1 / 0.49	1.9 / 0.44	1.9 / 0.37	8×10^{-6}
20,000	4.1 / 1.28	6.4 / 0.63	4.0 / 0.54	3.9 / 0.40	1×10^{-5}
40,000	8.2 / 2.28	12.8 / 0.88	7.4 / 0.75	7.4 / 0.48	2×10^{-5}
80,000	17.9 / 3.47	26.9 / 1.32	16.3 / 1.18	16.0 / 0.61	2×10^{-5}
Bytes per genotype	4.06	1.26	1.14	0.36	

4.11 Case-control outcomes and sampling weights: prespecified validation

A prespecified simulation (archive `20261005-hratt-weights-binary`, protocol frozen before the main run) drew fresh Balding–Nichols populations of 40,000, samples of 5,000 under six selection scenarios with inverse-probability weights, 20,000 independent markers with half at base minor allele frequency 1–5%, null and mixed traits, and binary traits at 1, 5 and 20% prevalence: 30 replicates per null cell and 10 per mixed cell. Two of its seven criteria passed.

Weighted quantitative tests were calibrated without structure (Table 17): λ_{GC} 0.995–1.003 and 0.94–0.99 times the nominal rate at $\alpha = 10^{-3}$, including selection on the outcome (S3), where the Huber–White sandwich and LDAK's weighted regression rejected 12 and 10 times too often for minor allele frequency 1–5%. Weighted HRATT gained power over the same test without a polygenic offset, by mean QTL χ^2 ratios of 1.39 to 2.63. With structure and weights that depend on ancestry (S4), exchangeability failed: λ_{GC} was 0.93, and variants with minor allele frequency 1–5% were rejected 5.75 times too often at 10^{-3} and 17 times at 10^{-4}, where the sandwich was

calibrated. Ancestry covariates alone did not fix this; with covariate-specific genotype distributions they did (Section 4.12).

Binary tests had lopsided tails at 1% and 5% prevalence (Table 18). The saddlepoint p-value adds both tails at $\pm|u|$, as SAIGE's does, so a skewed null does not put $\alpha/2$ in each tail, which the per-tail criterion required. Two-sided rates were 0.94–1.03 times nominal at 10^{-3} and 0.85–1.22 at 10^{-4} in every cell but S4. Without the saddlepoint the heavier tail reached 46 times nominal at 10^{-4}, and the model-based logistic variance gave $\lambda_{\mathrm{GC}} = 0.36$ at 1% prevalence. Official LDAK-KVIK was conservative at 1% prevalence (two-sided 0.62 and 0.57). Two of 30 replicates at 1% prevalence stopped with a separation error from the per-group null fit with its in-sample offset; with cross-fitted scores none did (Section 4.12).

HRATT's effect estimates were attenuated in every cell: by 32% without selection and by 33–46% in S1–S3, against −4% to +0.3% for LDAK-KVIK and weighted least squares, with correlation 0.997 to the population slopes throughout. The in-sample LOCO prediction absorbs part of each QTL's effect. p-values are unaffected. Cross-fitted LOCO scores removed the attenuation (Section 4.12). Excluding the saddlepoint, the weighted paths took 1.32 and 1.28 times as long as HRATT-HE, against prespecified limits of 1.3 and 1.6; the saddlepoint took a median 1.2–1.7 seconds per fit, about as long as an HRATT-HE fit.

Table 17: Weighted quantitative tests: rejection rates of null variants over nominal, pooled over 30 replicates, with λ_{GC} on minor allele frequency $\geq 5\%$. HRATT is the retrospective test with saddlepoint tails; "normal tails" and "sandwich (HC0)" replace its tails or its variance on the same residuals. LDAK-KVIK is unweighted (S0); "LDAK sandwich" is LDAK's weighted linear regression.

			$\alpha = 10^{-3}$, MAF		10^{-4}
Scenario	Method	λ_{GC}	1–5%	$\geq 5\%$	all
S0, $F_{ST} = 0$	HRATT	1.001	0.97	0.94	0.70
	normal tails	1.001	1.03	0.94	0.72
	sandwich (HC0)	1.003	0.90	0.94	0.64
	LDAK-KVIK	1.002	1.03	0.95	0.77
S1, $F_{ST} = 0$	HRATT	1.001	1.05	0.94	0.77
	normal tails	1.001	1.15	0.95	0.89
	sandwich (HC0)	1.004	0.92	0.92	0.75
	LDAK sandwich	1.004	1.37	0.98	1.41
S2, $F_{ST} = 0$	HRATT	0.995	0.91	1.00	0.84
	normal tails	0.995	1.36	0.99	1.32
	sandwich (HC0)	1.010	0.62	1.00	0.64
	LDAK sandwich	1.017	1.47	1.16	1.74
S3, $F_{ST} = 0$	HRATT	1.003	0.90	0.98	0.92
	normal tails	1.003	3.28	1.07	5.66
	sandwich (HC0)	1.037	12	1.68	26
	LDAK sandwich	1.067	9.98	1.68	18
S0, $F_{ST} = 0.05$	HRATT	0.978	0.76	0.88	0.89
	normal tails	0.978	0.85	0.88	0.90
	sandwich (HC0)	0.978	0.79	0.90	0.87
	LDAK-KVIK	0.997	0.91	0.98	1.05
S1, $F_{ST} = 0.05$	HRATT	0.973	0.98	0.81	0.79
	normal tails	0.973	1.14	0.81	0.85
	sandwich (HC0)	0.976	0.78	0.79	0.66
	LDAK sandwich	1.034	1.44	1.15	1.35
S4, $F_{ST} = 0.05$	HRATT	0.932	5.75	1.30	7.38
	normal tails	0.932	6.20	1.31	8.09
	sandwich (HC0)	0.979	0.78	0.79	0.69
	LDAK sandwich	1.033	1.35	1.14	1.38

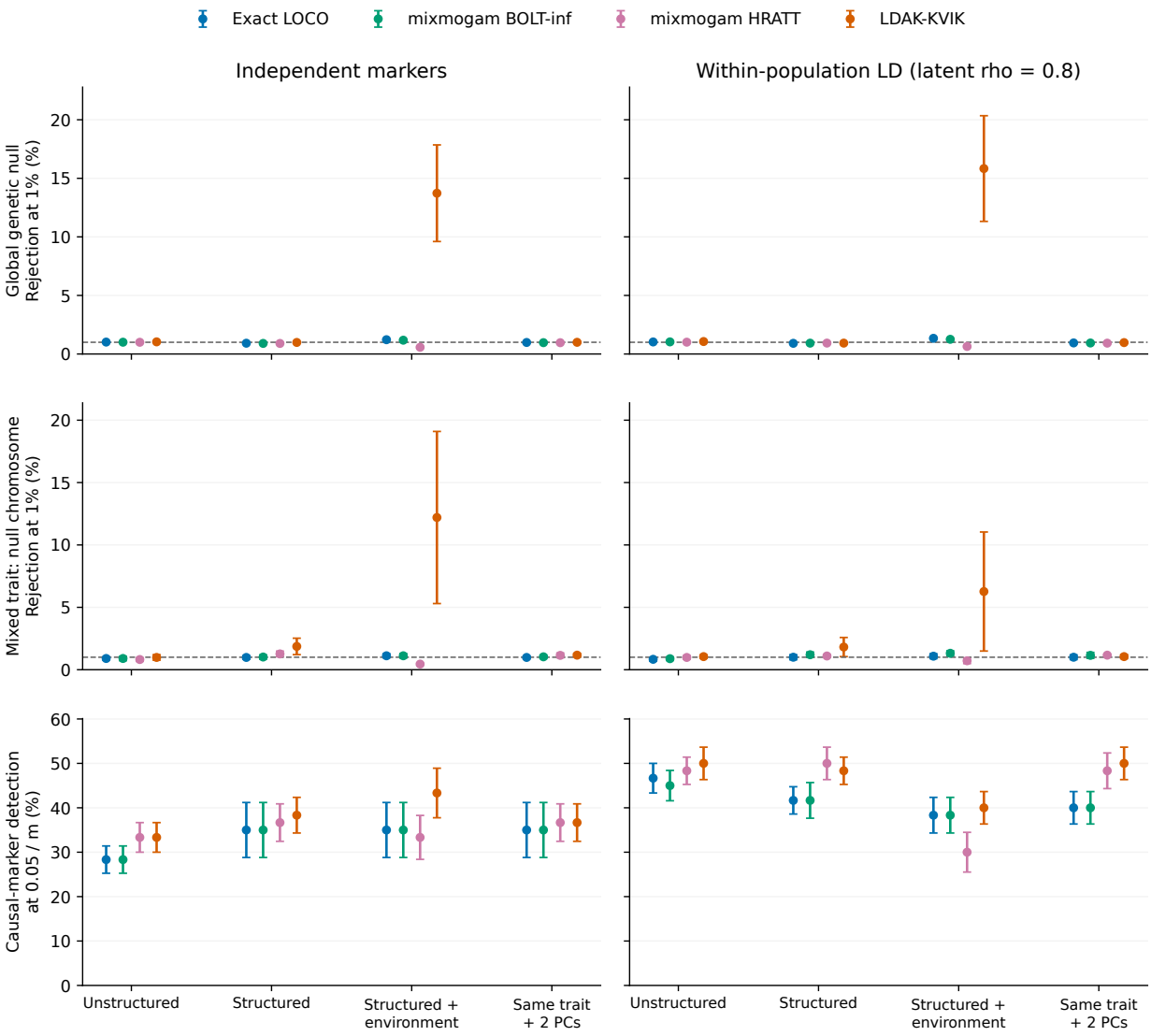


Figure 2: Matched phensim comparison at 800 samples and 6,000 markers; mixmogam methods are reruns with 2.0.0.dev6 on the original inputs. Bars show one Monte Carlo standard error across six independent panels. Dashed lines indicate nominal 1% rejection. The middle row uses only the chromosome without generating effects; the last row measures ten causal markers at $0.05/m$. Omitted environmental effects create associations under the genetic null. Detection in inflated cells is not calibrated power. Columns share scales within each row.

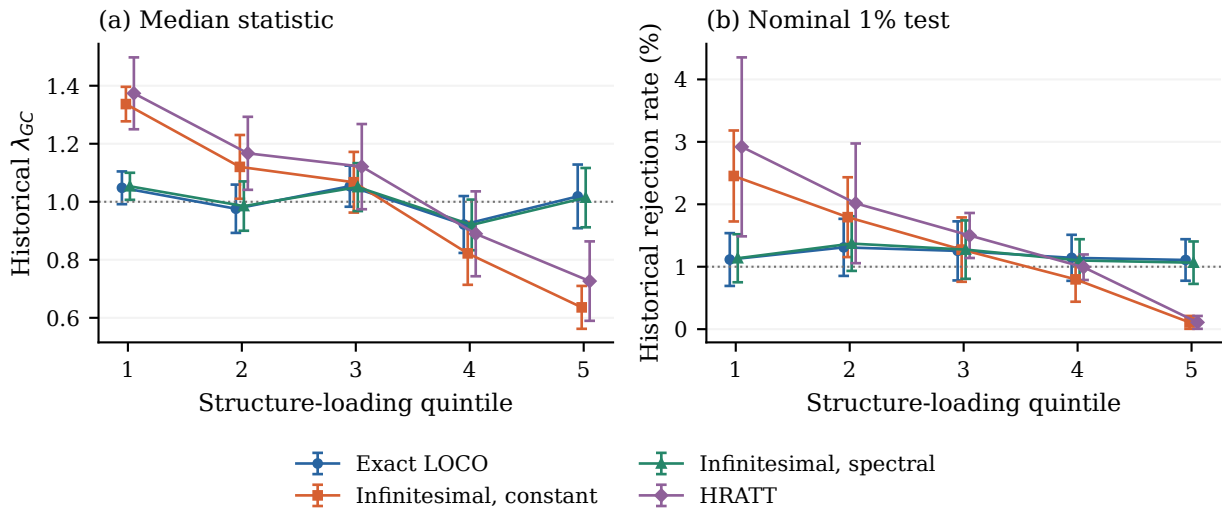


Figure 3: Historical strong-structure diagnostics. Points are means over six phenotype replicates on one fixed genotype panel; bars are 95% Student-t intervals for those means. The median statistic and the nominal 1% rejection fraction show why reporting only a genome-wide summary can obscure differences among variant groups. No external program results are included.

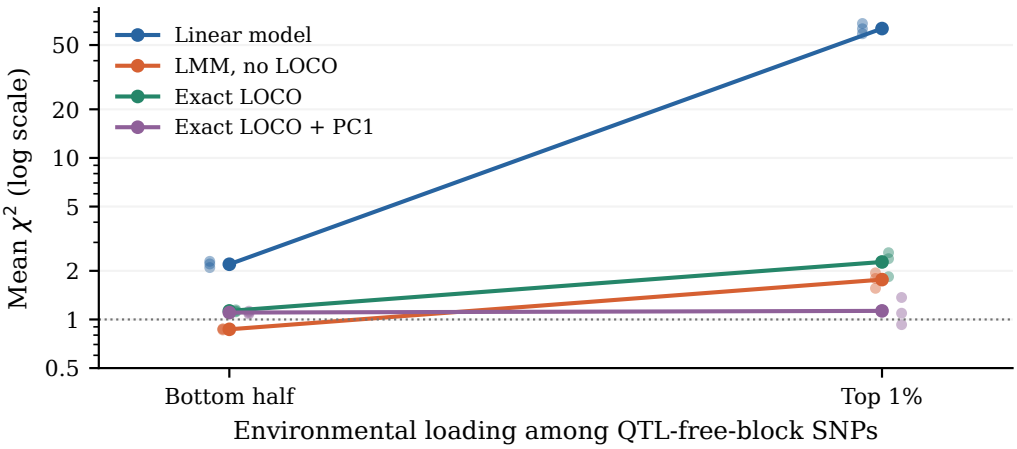


Figure 4: Historical S2 statistics by environmental loading in QTL-free blocks. Large points and lines show means; small points show the three simulation seeds. A logarithmic vertical axis accommodates the large unadjusted mean. PC adjustment reduces the observed gradient.

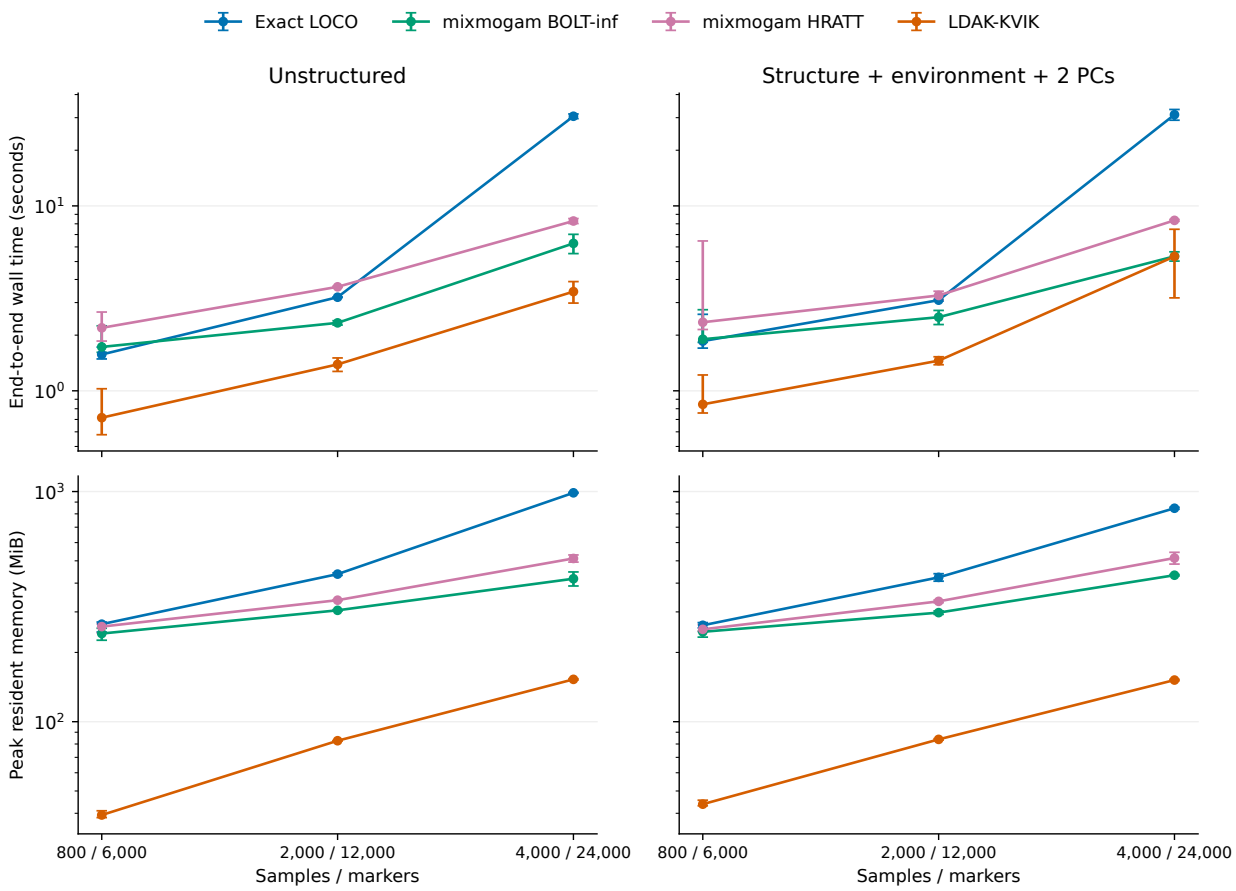


Figure 5: End-to-end time and process peak RSS across three controlled workloads. Points are medians; bars span the observed range across six replicates at n=800 and two at n=2,000 or 4,000. Cost axes are logarithmic. Sample and marker counts increase together. mixmogam points are 2.0.0.dev6 reruns; LDAK-KVIK points are the original measurements.

Table 18: Binary tests: rejection rates of null variants with minor allele frequency $\geq 1\%$ in each tail, over $\alpha/2$, by the sign of the effect, pooled over 30 replicates; λ_{GC} on minor allele frequency $\geq 5\%$. Cells give prevalence and scenario. "Model-based variance" replaces the retrospective variance by $\sum_i \mu_i(1-\mu_i)\tilde{g}_{ij}^2$ with normal tails; LDAK-KVIK is unweighted.

			$\alpha = 10^{-3}$		$\alpha = 10^{-4}$	
Cell	Method	λ_{GC}	Upper	Lower	Upper	Lower
1%, S0	HRATT	1.032	0.14	1.84	0.32	2.12
	normal tails	1.032	0.14	11	0.29	46
	model-based variance	0.357	0.05	0.98	0.11	1.79
	LDAK-KVIK	1.077	0.36	0.89	0.32	0.83
5%, S0	HRATT	1.006	0.43	1.53	0.33	1.37
	normal tails	1.006	0.45	1.97	0.37	2.91
	model-based variance	0.884	0.23	1.03	0.10	1.37
	LDAK-KVIK	1.031	0.91	0.91	0.74	0.97
20%, S0	HRATT	0.994	0.74	1.25	0.54	1.17
	normal tails	0.994	0.76	1.30	0.64	1.37
	model-based variance	0.965	0.63	1.05	0.54	1.17
	LDAK-KVIK	1.000	1.00	1.03	1.07	0.90
5%, S1	HRATT, weighted	1.007	0.34	1.69	0.30	1.71
	normal tails	1.007	0.36	2.51	0.33	4.39
5%, S2	HRATT, weighted	1.011	0.37	1.69	0.37	1.44
	normal tails	1.011	0.37	3.44	0.40	8.41
5%, S3	HRATT, weighted	1.003	0.64	1.36	0.50	1.47
	normal tails	1.003	0.66	1.42	0.57	1.68
5%, S5 1:1	HRATT, weighted	1.002	1.20	0.71	1.17	0.70
	normal tails	1.002	1.27	0.73	1.41	0.74
	HRATT, unweighted	1.001	0.93	0.94	1.11	0.74
	LDAK-KVIK	1.002	0.97	0.97	1.11	0.84
5%, S5 1:4	HRATT, weighted	1.002	0.81	1.21	0.70	1.37
	normal tails	1.002	0.83	1.24	0.70	1.64
	HRATT, unweighted	1.001	0.73	1.30	0.57	1.78
	LDAK-KVIK	1.020	0.97	1.03	1.17	1.31
5%, S0, $F_{ST} = 0.05$	HRATT	1.002	0.47	1.53	0.30	1.70
	normal tails	1.002	0.48	2.00	0.33	3.50
	model-based variance	0.880	0.27	1.13	0.15	1.77
5%, S4, $F_{ST} = 0.05$	HRATT, weighted	0.933	1.15	2.59	0.89	4.96
	normal tails	0.933	1.93	4.00	2.80	12

4.12 Cross-fitted scores and covariate-specific genotype distributions: prespecified validation

The follow-ups of Section 2.13 were validated on the samples of Section 4.11, regenerated from their seeds (archive `20261006-hratt-followups`, protocol frozen before the main run). The new cells were mixed traits at $F_{ST} = 0.05$ and variants of the structured cells with 10 principal components among the covariates, which the driver computed from 2,000 thinned markers by four iterations of a randomized subspace method. All 660 cases completed, and five of seven criteria passed.

Effects. Cross-fitting removed the attenuation (Table 19):

- Without structure or selection on the outcome (S0–S2), HRATT's effects were within 1% of the population slopes. Unweighted, it was 8% low under selection on the outcome, as the design expects.
- At $F_{ST} = 0.05$ without principal components they were 2.0% below the within-population slopes.
- In-sample scores gave -28 to -36%.
- Under selection on the outcome (S3), weighted HRATT was 5.4% low (Monte Carlo SE 2.1), failing that criterion. Weighted least squares was 4.0% low on the same samples, so inverse-probability weighting, not HRATT, accounts for most of the gap.
- Binary one-step effects were within 3% of the population marginal log odds ratios, with Monte Carlo SEs of 4–9%; in-sample scores gave -16% and -18%.

Power did not change: mean QTL χ^2 was 0.999–1.009 times that of in-sample scores. Weighted HRATT exceeded weighted least squares by 1.20 to 1.29 times, and binary HRATT exceeded logistic regression by 1.04 times.

Calibration. Without ancestry-dependent weights every test stayed calibrated (Table 20):

- Quantitative λ_{GC} was 0.993–1.008, with 0.92–1.04 times the nominal rate at 10^{-3}.
- Binary two-sided rates were 0.94–1.08 at 10^{-3} and 0.84–1.14 at 10^{-4}. At 1% prevalence no replicate failed (2 of 30 on 5 October).
- The doubled tails put about half of $\alpha/2$ in the upper tail at 1% prevalence (0.51 at 10^{-3}), at the edge of the criterion's range.
- On chromosome 10 of the mixed traits λ_{GC} was 0.971–1.024. This includes the refitted scores with λ at $F_{ST} = 0.05$.

With weights that depend on ancestry (S4) and principal components among the covariates, weighted tests rejected low-frequency variants at 0.85 (quantitative) and 0.96 (binary) times the nominal rate at 10^{-3}. The pooled-frequency model gave 6.54 and 3.96 with the same covariates, and the new model without principal components 5.86 and 3.16.

A rerun of the five S4 cells on 8 October (archive `20261008-hratt-followups-builtin-pcs`, 130 cases, protocol frozen before the run) replaced those components by the built-in exact SVD of the same thinned panels; cells without components matched the 6 October results to three decimals. The rates became 0.82 and 0.98 times nominal, with λ_{GC} 0.989 and 0.984, against 6.64 and 3.95

with pooled frequencies; in the 10-replicate mixed cell the rate was 0.77 against 0.92. The effect, power and chromosome-10 criteria still passed on these cells. One binary case crashed in Apple Accelerate's `dgemm` on a loaded host and completed cleanly when regenerated; its cell uses the other 29 replicates.

Time. Cross-fitting took 1.11 (quantitative) and 1.08 (binary) times as long as in-sample scores on the same cases. Weighted HRATT took 1.55 times as long as HRATT-HE, excluding the saddlepoint, failing the limit of 1.3; it was 1.32 on 5 October, before the covariate-specific frequencies.

4.13 Known major genes on the RegMap panel: prespecified check

Simulated data cannot contain the errors of real files, such as sample IDs that do not join or positions on another assembly. The bundled RegMap panel (1,307 accessions, 214,051 SNPs) carries the phenotypes of Atwell et al. (2010). For seven of them a single gene with a large effect is known (Table 21). Before the first fit we fixed the controls, raw phenotype values, a window of 50 kb around each TAIR10 gene body, the threshold $0.05/m$ and two criteria (archive `20261017-atwell-controls`). A1 required exact LOCO to find all seven genes; it failed. A2 required BOLT-LMM-inf and HRATT to find every gene exact LOCO found; it passed.

All four methods found the same six genes. The best window variant ranked first or second genome-wide, inside the gene body or within 7 kb of it. Every phenotyped accession joined the panel, and the signals sit at the genes, so the sample join and the positions hold on these files. (Allele coding cannot be checked this way: swapping it leaves p unchanged.) No method found MOT1 for molybdenum. In an exploratory rerun on $\log_{10}$ values, exact LOCO ranked a variant inside MOT1 first ($p = 6 \times 10^{-10}$). Seven accessions carry its minor allele, with median Mo 0.47 against 2.53 for the rest. Baxter et al. (2008) traced low shoot Mo to a MOT1 promoter deletion in seven low-Mo accessions; we did not check whether they are the same seven. On the raw scale the high tail of the common allele dominates the variance. The miss lies in the prespecified raw scale, and a real-data plan should fix each trait's scale in advance.

HRATT's variational fits did not converge on the five traits where its heritability estimate reached the upper bound (0.992 or more): the four hypersensitive responses and FLC expression. They ran to 100 iterations in 101–143 seconds, against 12–30 seconds for the converged fits and under two seconds for the other methods. They still found every control. These samples hold 84 to 166 accessions, where `auto` uses exact LOCO. The check covers data handling and power at large effects.

Table 19: Effects at the QTL of mixed traits: relative bias (%) of the per-allele estimates against the population slopes (within-population at $F_{ST} = 0.05$; marginal log odds ratios for binary traits), mean over 10 replicates with its Monte Carlo SE, and mean QTL χ^2.

Cell	Method	Bias (SE)	QTL χ^2
S0	HRATT	−0.7 (0.7)	152.6
	in-sample scores	−32.4 (1.2)	151.6
	weighted HRATT	−0.6 (0.7)	150.9
	weighted LS	−0.8 (0.7)	124.2
S1	HRATT	+0.9 (0.8)	154.0
	in-sample scores	−33.5 (1.5)	152.9
	weighted HRATT	+0.3 (0.8)	116.2
	weighted LS	+0.0 (1.0)	96.8
S2	HRATT	−0.6 (1.0)	149.6
	in-sample scores	−33.0 (1.9)	149.1
	weighted HRATT	−1.0 (1.5)	73.8
	weighted LS	−1.3 (1.3)	60.1
S3	HRATT	−8.2 (1.1)	138.6
	in-sample scores	−36.0 (1.5)	138.2
	weighted HRATT	−5.4 (2.1)	41.6
	weighted LS	−4.0 (2.9)	32.3
S0, $F_{ST} = 0.05$	HRATT	−2.0 (1.9)	145.5
	in-sample scores	−28.0 (2.1)	145.7
	weighted HRATT	−5.5 (2.6)	138.2
	weighted LS	−3.5 (2.3)	125.3
S0, $F_{ST} = 0.05$, PCs	HRATT	−1.0 (1.2)	148.5
	in-sample scores	−33.3 (1.9)	147.4
	weighted HRATT	−1.0 (1.2)	145.7
	weighted LS	−0.4 (1.3)	122.2
S4, $F_{ST} = 0.05$, PCs	HRATT	−0.7 (0.7)	149.1
	in-sample scores	−32.7 (1.3)	148.7
	weighted HRATT	−0.3 (1.0)	115.2
	weighted LS	−0.2 (1.2)	96.8
5%, S0	HRATT	+1.0 (9.0)	24.6
	in-sample scores	−15.8 (5.7)	24.6
	weighted HRATT	+0.9 (9.0)	24.6
	logistic	−2.6 (8.8)	23.8
5%, S5 1:4	HRATT	−3.0 (4.4)	75.2
	in-sample scores	−17.6 (2.5)	75.5
	weighted HRATT	+1.2 (9.4)	70.8
	weighted logistic	−1.1 (9.1)	71.4

Table 20: Calibration of the follow-up tests: rejection rates of null variants over nominal, pooled over 30 replicates, with λ_{GC} on minor allele frequency $\geq 5\%$. "Doubled tails" doubles the saddlepoint tail beyond the score; "pooled frequencies" gives every sample the pooled allele frequency.

			$\alpha = 10^{-3}$, MAF		10^{-4}
Cell	Method	λ_{GC}	1–5%	$\geq 5\%$	all
S0	HRATT	1.002	1.01	0.93	0.74
	weighted	1.002	0.96	0.94	0.72
S1	HRATT	0.993	1.11	0.97	0.97
	weighted	1.002	1.03	0.94	0.84
S2	HRATT	1.008	0.95	0.97	0.94
	weighted	0.993	0.85	0.99	0.77
S3	HRATT	0.999	0.99	1.00	1.09
	weighted	1.006	0.97	1.02	0.99
1%, S0	HRATT	1.012	1.01	1.06	1.12
	doubled tails	1.012	0.52	1.01	0.80
5%, S0	HRATT	1.003	0.98	0.97	0.85
	doubled tails	1.003	0.83	0.94	0.84
20%, S0	HRATT	0.995	0.96	1.04	0.94
	doubled tails	0.995	0.92	1.04	0.92
5%, S1	weighted	1.006	1.07	0.95	1.09
	doubled tails	1.006	0.98	1.02	0.95
5%, S2	weighted	1.009	1.09	1.07	0.95
	doubled tails	1.009	0.93	1.07	0.94
5%, S3	weighted	1.003	1.04	0.94	0.99
	doubled tails	1.003	0.98	0.93	0.87
5%, S5 1:1	weighted	1.004	0.92	0.95	0.92
	doubled tails	1.004	0.91	0.96	0.94
5%, S5 1:4	weighted	1.003	0.96	1.03	1.06
	doubled tails	1.003	0.97	1.04	1.00
S0, $F_{ST} = 0.05$, PCs	weighted	1.002	0.87	1.02	1.09
S1, $F_{ST} = 0.05$, PCs	weighted	1.004	1.08	0.93	0.98
S4, $F_{ST} = 0.05$, PCs	weighted	0.990	0.85	0.93	0.86
	pooled frequencies	0.941	6.54	1.47	9.09
S4, $F_{ST} = 0.05$	weighted, no PCs	0.931	5.86	1.29	7.49
5%, S0, $F_{ST} = 0.05$, PCs	weighted	1.001	0.99	0.91	0.92
5%, S4, $F_{ST} = 0.05$, PCs	weighted	0.985	0.96	0.91	0.82
	pooled frequencies	0.938	3.96	1.29	4.32
5%, S4, $F_{ST} = 0.05$	weighted, no PCs	0.914	3.16	1.06	3.08

Table 21: Known major genes on the RegMap panel: the smallest p within 50 kb of the gene body and, in parentheses, its genome-wide rank, for raw phenotypes. Exact is exact LOCO; OLS+PCs has 10 built-in principal components. †Above $0.05/m$ (2.4–2.5×10^{-7}). *No criterion: FLC and FRI both regulate FLC expression.

Trait	Gene	n	Exact	BOLT-inf	HRATT	OLS+PCs
avrPphB	RPS5	90	2×10^{-22} (1)	3×10^{-13} (1)	5×10^{-11} (1)	6×10^{-18} (1)
avrRpm1	RPM1	84	8×10^{-16} (1)	2×10^{-10} (2)	2×10^{-12} (1)	3×10^{-15} (1)
avrRpt2	RPS2	89	2×10^{-16} (1)	7×10^{-12} (1)	1×10^{-11} (1)	3×10^{-15} (1)
avrB	RPM1	87	7×10^{-13} (2)	3×10^{-9} (2)	1×10^{-9} (1)	8×10^{-13} (1)
FRI expression	FRI	163	3×10^{-10} (1)	8×10^{-9} (1)	2×10^{-8} (1)	1×10^{-11} (1)
Na	HKT1	93	4×10^{-10} (1)	6×10^{-8} (2)	7×10^{-9} (1)	5×10^{-10} (1)
Mo	MOT1	93	3×10^{-6} (6)†	1×10^{-5} (12)†	1×10^{-6} (10)†	2×10^{-6} (5)†
FLC expression*	FLC	166	2×10^{-3} (684)†	8×10^{-3} (2674)†	9×10^{-3} (1163)†	4×10^{-3} (2645)†
FLC expression*	FRI	166	5×10^{-3} (1513)†	1×10^{-2} (3835)†	6×10^{-3} (763)†	3×10^{-3} (2153)†

4.14 Sample-size scaling, cache budgets and convergence

On 29 October we reran the package benchmarks with 2.0.0.dev8 at commit `d550c412`, on an Apple M2 Pro with 16 GiB RAM. Selected measurements, manifests and diagnostics are retained in the compact `20261029-report-evidence` bundle; full inputs and worker output remain in the original local archives. The requested worker and native-library limits were one except where stated; the active Apple Accelerate thread pool could not be independently observed. Whole-process time includes startup and input handling. LDAK time sums its two processes, and its RSS is the larger process peak. The sample-size and matched-genome drivers observed load but did not forward their load limit to fit workers.

Table 22 fixes the marker count at 20,000 and varies the number of samples. Each entry is one run on a synthetic mixed trait. The panels share a 100-marker founder-block design but are distinct realizations, and the number of founders changes with sample size. HRATT-HE is the fastest tested method from 10,000 to 100,000 samples. At 100,000, three further balanced-order repetitions on the same input gave median [minimum, maximum] seconds of 25.836 [25.565, 26.307] for HRATT-HE, 37.589 [37.188, 37.610] for LDAK-KVIK, and 101.770 [101.540, 102.395] for BOLT-inf. Median peak RSS was 2.491, 1.261 and 3.598 GiB, respectively. Each method's p-values exactly matched its own first run. These repetitions measure timing variability on one input.

Table 22: Whole-process wall seconds in the 20,000-marker sample-size ladder, one run per cell. The panels are different realizations.

Samples	LDAK-KVIK	HRATT-HE	HRATT-REML	BOLT-inf	Exact
1,000	1.80	2.57	2.78	2.03	2.08
2,500	1.91	2.77	3.19	2.60	9.45
5,000	2.91	3.27	4.13	4.00	–
10,000	4.72	4.62	6.10	7.21	–
20,000	8.98	6.16	9.77	15.21	–
50,000	19.14	13.25	22.94	45.10	–
100,000	37.14	25.75	47.07	102.20	–

At 100,000 samples HRATT-HE and LDAK-KVIK had Spearman correlation 0.980 and 98 of the top 100 variants in common; BOLT-inf and LDAK had 0.861 and 97. These estimators differ. The protocol both says there is no agreement threshold and lists a 0.9 correlation check; the BOLT-inf comparison misses that check. On a separate 50,000-sample HAPNEST panel with longer-range LD, single runs took 17.48 seconds for HRATT-HE, 19.51 for LDAK-KVIK and 76.25 for BOLT-inf. The resource driver did not retain mixmogam p-values, so per-variant agreement was not checked there.

Table 23 compares the previous and current HRATT-HE implementations on two 50,000-sample HAPNEST panels. Each entry uses three warmed, alternating-order pairs. At the default cache budget, the small timing differences do not establish a speedup. With a deliberately restricted 28,000,000-byte budget, median paired baseline/current ratios were 1.093 and 1.129. All saved arrays matched exactly within each pair; shared diagnostics passed their tolerance and both variational stages converged with matching iteration counts. Initial cross-fit allocations were partial, but the new implementation can expand the cache after the hold-out fit releases its allocation; final occupancy was not measured. The restricted-cache run used a corrected archived benchmark driver, because the repository driver passes the cache option under the wrong keyword. It used the default cross-fitting configuration.

At 5,000 samples and 800,000 variants, one HRATT-HE fit per cache budget took 48.7 seconds

Table 23: Three paired runs per panel at 50,000 samples and 20,000 markers. Default-budget entries are median wall seconds. The restricted column is the median paired baseline/current wall-time ratio.

Panel	Baseline, s	dev8, s	Restricted ratio
Unstructured	18.950	18.665	1.093
Confounded, with PCs	16.761	16.686	1.129

with 1 GB and 47.1 with 2 GB; p-values matched exactly and both stages converged. The 1 GB budget initially retained 4,570 of 6,260 cross-fit blocks. Final occupancy was not measured, and two single timings cannot show a cache-budget speedup.

The 308-fit scale study includes ten independent replicates for each of five methods in each of five calibration cells. In the unstructured pure null, mean fractions below $p = 0.05$ were 0.0509–0.0512 across methods; in the structured null they were 0.0417–0.0483. BOLT-inf's structured-null fractions ranged from 0.0279 to 0.0703 across replicates. Variant-level binomial intervals in the original summary assume independence despite LD and do not describe uncertainty in the overall rate. In the confounded cell, BOLT mixture, HRATT-HE and HRATT-REML each had at least one non-converged stage in 10/10 replicates; with two PCs the counts were 7/10, 8/10 and 7/10. The aggregate summaries include those fits.

All finite masks matched in the configuration study. At 5,000 samples and 400,000 variants, unpinned-BLAS BOLT mixture exceeded the 10^{-6} absolute log-p tolerance at 8 of 400,000 variants, with maximum 8.39×10^{-6}; other configurations met it. A paused worker's original parent wall time includes the pause and is not a runtime measurement; an isolated repeat matched its stored arrays exactly.

The Atwell control rerun again recovered six of seven controls with every method: A1 failed on Mo98/MOT1 and A2 passed because HRATT and BOLT-inf recovered all controls found by exact LOCO. Four of eight HRATT scans had a non-converged stage; A2 does not exclude such fits. The earlier archive did not record comparable convergence fields, so we cannot tell whether convergence got worse.

5 Discussion

5.1 The central distinction is between geometry and inference

Equation (9) explains the conditional gradient without any software bug: covariance precision weights variant directions differently, and scalar calibration discards that. The known-covariance experiment shows that restoring a few leading directions removes most of the error when structure is concentrated in them and the rest of the spectrum is benign. It also shows that a good average rejection fraction is not enough.

The scope is deliberately narrow. The leading subspace is known exactly, the covariance is fixed, genotypes have no local LD, and the statistic is linear in a Gaussian phenotype before squaring. Estimated covariance, approximate eigenvectors, rare alleles, nonlinear shrinkage and phenotype-dependent selection each raise questions this calculation does not answer. The result justifies further work on the spectral correction; it does not make it a calibrated replacement for established GWAS procedures.

5.2 Prediction accuracy and test validity need separate targets

A flexible polygenic predictor can reduce residual variance and so improve association power, but it also changes the null distribution of a test built on its residuals. Calibration inherited from an infinitesimal model may not survive a nonlinear prior, tuning, or incomplete optimization. LD Score regression separates components of genome-wide inflation under its assumptions (Bulik-Sullivan et al., 2015), but an intercept-based rescaling says nothing about individual variants in a structured panel.

The same applies to prediction. An internally tuned or in-sample BLUP is not a prospective validation, which needs held-out samples that respect family relationships, ancestry and the target population. A model can predict better while its association statistic is miscalibrated, and a calibrated test can lose power through a poor background model.

5.3 Environmental confounding needs explicit definitions

The environmental experiment illustrates the noncentral mean term in Equation (12). An exposure generated from deme membership is causally external to the candidate variants, but statistically dependent on them; that dependence is the confounding being studied. A PC computed from the tested genotypes is also a legitimate stress axis, but it answers a different question from a separately generated exposure. Future studies should include both and state the causal and statistical role of each.

5.4 Limitations and intended use

mixmogam is best used as a transparent research implementation and a framework for controlled comparisons. The main limitations of the evidence are these.

- The conditional experiment uses three small fixed panels, a known covariance and exact eigenvectors. It does not exercise the production randomized eigensolver, its width selection, or phenotype-dependent choice of calibration markers.
- The archived simulations have three or six replicates and predate important fixes.
- The matched LDAK-KVIK comparison has six panels per cell at $n = 800$, AR(1) rather than demographic LD, and common genetic effects across three discrete populations. The two programs fit different models, the official timings come from other days, and the Mac LDAK build lacks MKL and OpenMP.
- The larger workloads are single synthetic realizations on one workstation, often with one phenotype. Repeated timings measure computational variability, not variation across populations.
- There is no validation in human multi-ancestry cohorts, at biobank scale, or with imputed dosages; the input contract is hard calls.
- Case-control outcomes and sampling weights are supported only by HRATT (Section 2.12) and were tested in simulations of unrelated samples with independent markers (Sections 4.11 and 4.12). Weighted effects under selection on the outcome stayed 5.4% low and weighted fits took 1.55 times as long as HRATT-HE, both failing their criteria. The saddlepoint assumes Hardy–Weinberg equilibrium given the covariates and, being retrospective, differs from SAIGE's model-based one (Zhou et al., 2018). Rarer outcomes, relatedness and real participation weights are untested.

- Variational fits did not always converge: under strong confounding in the scale study, and in four of eight Atwell HRATT scans. These fits are included in the summaries and in the A2 criterion.
- Four-thread fits at 20,000 markers fail a strict array tolerance against one thread, although tested decisions are unchanged.
- The RegMap check tests data handling and power for large effects, not tail calibration.

The package's modes are not interchangeable. Dense plug-in tests, iterative infinitesimal tests and nonlinear two-step heuristics approximate different things, so a result needs its mode, grouping, covariance estimation, convergence diagnostics and genotype definition reported with it. Likewise, reusing work, choosing a cheaper variance estimator and changing storage are distinct interventions: the first can preserve a result, the second changes the fitted model, and the third trades time for memory. Their gains should not be folded into a single speedup. REML (without sampling weights) and one thread remain the defaults.

6 Future research: hypotheses and decisive experiments

The priorities are ordered by dependence: reliable input identity and reproducible reference comparisons come before claims about a new method's power or speed.

6.1 Priority 1: broaden the matched-input reference benchmark

Hypothesis. With input handling and nuisance definitions matched, the local implementations reproduce an independent Gaussian reference, and the remaining differences from external programs can be explained.

Experiment. Keep the input contract of Section 3.2 and add missing calls, admixed and related samples, demographic LD, larger panels and more replicates, with official BOLT-LMM, REGENIE and Quickdraws alongside LDAK-KVIK. Generate large cohorts from a HAPNEST-model reference tuned on one subset and audited (frequencies, LD decay, PCs, donor sharing) on another, varying sample size and marker density separately. Cross projected HE and REML with polygenic and sparse architectures and with known or omitted environmental means; LDAK's partitioned estimator and its REML revision need their own reference configurations. Discrepancies under an identical Gaussian model must be explained at the effect, standard-error, variance-component and statistic levels before aggregate concordance is accepted.

6.2 Priority 2: test calibration independently of adaptive choices

Hypothesis. An approximation selected on calibration markers keeps its accuracy on independent audit markers and genotype panels.

Experiment. Choose spectral width and scaling on one set of markers (whole LD blocks where possible) and audit on another, repeating any phenotype-dependent selection for every null phenotype. Stratify by frequency, missingness, LD score, ancestry loading and kinship leverage, first under a known covariance and then with fitted variance components. Fix the acceptable rejection error at each threshold in advance, for example a ratio within 0.9–1.1 at $\alpha = 0.01$ in every adequately powered stratum, and give genome-wide tails their own error budget. Persistent failure in a stratum should lead to a more accurate denominator or an explicit unsupported-case result.

6.3 Priority 3: replace spectral heuristics with computable error bounds

Hypothesis. A hybrid method can keep streaming efficiency while bounding denominator error for difficult variants.

If the omitted eigenvalues lie in $[a, b]$ with $a + \delta > 0$ and $s_j = \|\mathbf{z}_j\|^2 - \sum_{\ell \le k} a_{\ell j}^2$, the exact retained contribution $q_{j,\le k}$ gives

$$q_{j,\le k} + \frac{s_j}{b+\delta} \;\le\; q_j \;\le\; q_{j,\le k} + \frac{s_j}{a+\delta}. \tag{19}$$

Propagated through Equation (10), this interval says whether the approximation is adequate at the requested threshold; variants with wide intervals could receive an exact iterative solve. For $A = \mathbf{K} + \delta\mathbf{I} \succeq \delta\mathbf{I}$, an approximate solution $\widehat{\mathbf{w}}$ and residual $\mathbf{r} = \mathbf{y} - A\widehat{\mathbf{w}}$,

$$\left|\mathbf{z}_j^\top (A^{-1}\mathbf{y} - \widehat{\mathbf{w}})\right| \le \frac{\|\mathbf{z}_j\|\,\|\mathbf{r}\|}{\delta} \tag{20}$$

bounds the numerator error in the same way. Neither bound is implemented, and Ritz values alone do not certify the spectrum.

Experiment. Compare the bounds with dense references on small problems, then record how often fallback solves are needed, peak memory and end-to-end time. The approach is worth keeping only if the bounds are valid and selective enough to save work.

6.4 Priority 4: derive calibration for nonlinear polygenic residuals

Hypothesis. Mixture and elastic-net residuals need a calibration that reflects their own response to the phenotype, not a transferred infinitesimal factor.

Experiment. Step through fixed ridge, tuned ridge, a fixed mixture prior and jointly tuned nonlinear models. For each, compare an analytic or influence-function approximation with a parametric bootstrap that repeats fitting and hyperparameter selection, under correctly and deliberately misspecified effect distributions. A power gain counts only at matched type-I error and once separated from incomplete optimization and reuse of data for tuning.

6.5 Priority 5: separate mean confounding, covariance error, and localization

Hypothesis. The benefit and cost of PC adjustment depend on which part of the data-generating process is misspecified, and apparent null-locus inflation depends on the discovery unit.

Experiment. A factorial simulation varying ancestry-associated exposure from zero to the S2 stress level, covariance from known to fitted and misspecified kinships, and covariates from none to measured exposure and prespecified PCs, across relatives, admixture, unequal ancestry sizes and LD. Report effect bias, coverage, conditional rejection, calibrated power and between-ancestry performance, and compare fixed, LD-aware and simulated-independent blocks on the same genotypes. If LOCO is calibrated without PCs at modest exposure, the S2 result stays a stress-test finding; if PCs remove genuine ancestry-dependent signal, that trade-off must be reported rather than tuning λ_{GC} to one.

6.6 Priority 6: make scale and reproducibility part of method design

Hypothesis. Compact hard-call access and better placement of decoding work can close the remaining memory gap without losing the speed gained by reusing matrix products.

Experiment. Repeat both programs in the 20,000-marker comparison on the same day on an idle host, then against the pinned Linux OpenMP/MKL build, with balanced order, CPU time and complete-fit boundaries. Vary sample size and marker density separately and report peak memory per genotype for int8 and two-bit calls, with setup, variance fitting, prediction and testing timed apart; REML and BOLT-LMM still take 1.14 to 1.23 times as long as with dev5's cache. Before accepting a looser numerical tolerance, test high LD, small residual variance and phenotypes near decision thresholds. HRATT's case-control and weighted tests need external validation at prevalence below 1%, with related samples, real participation weights and SAIGE and REGENIE as comparators. A scalability claim should state the validated data size, hardware, resource limits and failure boundary.

Data, software, and reproducibility

The repository is available at https://github.com/bvilhjal/mixmogam. Earlier report sections describe package version 2.0.0.dev6 and the unreleased changes after it; the new benchmark section identifies 2.0.0.dev8 and its commit. The known-covariance experiment uses baseline commit `729c8532`. Its self-contained driver is `benchmarks/denominator_geometry.py`, with results, source snapshot, and manifest in `20261003-manuscript-geometry` beneath `benchmarks/results/`. The review archive is `20261003-critical-review`. Historical inputs retain their original source snapshots. The excluded external-program archive is also preserved for provenance; its results are not incorporated into the revised displays.

Analyses run from Python or from a thin command line: `mixmogam gwas` reads PLINK genotypes with phenotype, covariate and sampling-weight files (covariates in the PLINK `.cov` or a single-ID layout, aligned by sample ID), optionally appends the built-in principal components, and writes summary statistics; `mixmogam pca` writes the components as a covariate file. Both outputs are byte-identical to the same call through the Python interface, which tests check. The CSV's `af` counts the effect allele (PLINK BIM A1), not the minor allele. Ordinary quantitative scans export the headcount `n`; binary and sampling-weighted scores omit it to prevent its interpretation as an effective size. An explicitly supplied `n_eff` remains separate. A declared `genome_build` survives export and variant selection without converting coordinates or validating the assembly. `docs/tutorial.md` applies the methods to a simulated structured population.

The matched phensim study is archived as `20261003-phensim-kvik`, using mixmogam baseline `db232216` and phensim baseline `4939124e` with the explicitly snapshotted simulator extensions. The generating and analysis driver is `benchmarks/ldak_kvik_comparison.py`, archived under its earlier name `kvik_simulation.py`; the frozen plan, full package sources, complete inputs, raw p-values, commands, logs, diagnostics and resource measurements accompany its results. The separate `report/make_ldak_kvik_figures.py` renders this evidence without rerunning any association test.

The workload extensions are listed in the report evidence map. Large BED files remain local, with hashes, phased references and generating sources retained in their archives. HAPNEST resource summaries here link to full phensim source/reference archives. The simulation-only and association measurements must not be combined into a claim about the cost of a whole-genome analysis at 100,000 samples.

The implementation-only comparison is frozen in `20261003-kvik-efficiency`. The estimator comparison, `20261003-kvik-he-efficiency`, also retains the independent variance and global-null validation.

The 20,000-marker inputs and their all-call audit are archived as

20261003-hapnest-kvik-n50000-m20000.

The original thread and cache experiments are in 20261003-kvik-20k. They retain commands, raw measurements, association arrays, model diagnostics and verification records, including numerical failures and VM observations. Their frozen source version is 2.0.0.dev3.

The table generator, report/make_efficiency_tables.py, reads saved individual measurements and never reruns a model. Its efficiency_table_manifest.json records input and output hashes.

Two archives hold the 2.0.0.dev5 evidence. The paired archive, 20261004-efficiency-paired, comes from benchmarks/efficiency_paired.py and keeps both frozen sources, panel hashes, process measurements and saved results. The Lanczos archive, 20261004-slq-defaults, comes from benchmarks/slq_defaults.py and keeps every fit with its source snapshot. report/make_dev5_tables.py renders Table 13 from the paired archive and Table 14 from the Lanczos rerun, and records hashes in dev5_table_manifest.json.

The benchmark reruns with 2.0.0.dev6 are archived as 20261006-phensim-kvik-dev6 (with -n2000 and -n4000), 20261006-hapnest-kvik-dev6-n10000 and -n50000, 20261006-kvik-he-dev6 and 20261006-kvik-20k-dev6, beside the earlier reruns with commit e4089d8 (20261005-*-e4089d8) and 2.0.0.dev5 (20261004-*-dev5). Each records its frozen source, its commands and the hash of every reused file. The same-day checks are 20261004-same-day-dev4-dev5, 20261005-same-day-dev5-e4089d8 and 20261006-same-day-e4089d8-dev6. The figure and table generators read the reruns, take official LDAK-KVIK results from the original archives, and verify reused inputs and outputs by hash. The cross-fit rerun with threaded genotype products, which Table 10 reports, its same-day pairs and the probe behind the change are in 20261007-kvik-20k-crossfit-gemm.

The paired dev5/current archive is 20261005-efficiency-paired-hratt. benchmarks/efficiency_paired.py produced it, and report/make_streaming_tables.py renders Tables 15 and 16 from it. The steps between the versions are in 20261004-genotype-streaming, 20261004-gram-cache and 20261004-packed-calls. The Lanczos rerun is 20261005-slq-defaults-e4089d8.

The case-control and weighted validations are archived as 20261005-hratt-weights-binary and 20261006-hratt-followups, with the rerun on the built-in principal components in 20261008-hratt-followups-builtin-pcs, from benchmarks/hratt_weights_binary.py and benchmarks/hratt_followups.py; report/make_weights_binary_tables.py and report/make_followup_tables.py render their tables.

report/README.md gives the reproduction commands, exact archive identities, and evidence map. The figure generator reads recorded results and does not silently rerun old analyses. The figure and report manifests identify source hashes; the bibliography metadata records verified DOI identities. No new human-participant data were collected for this revision. The new experiments are simulated, apart from the positive-control check on the bundled RegMap panel (Section 4.13). Historical RegMap results are not used to support current performance claims.

A Useful identities and interpretation checks

A.1 Whitening explains the known-shape F reference

Let $L = \mathbf{H}^{-1/2}$ and let Q span $L\mathbf{X}$ orthonormally. The whitened residual phenotype is $\mathbf{r} = (\mathbf{I} - QQ^\top)L\mathbf{y}$ and the residual candidate is $\mathbf{v}_j = (\mathbf{I} - QQ^\top)L\mathbf{z}_j$. Under the known-shape Gaussian null, the squared projection onto $\mathbf{v}_j$ and the remaining residual sum of squares are independent scaled chi-squares with one and $n - p - 1$ degrees of freedom. Their ratio is Equation (7). When L

is estimated from $\mathbf{y}$, this fixed-linear- transformation argument no longer proves the reference law. It explains precisely where the plug-in approximation enters.

A.2 Residual coordinates for permutation

Choose an $n \times (n-p)$ orthonormal matrix B spanning the complement of $L\mathbf{X}$. With known Gaussian covariance shape,

$$\mathbf{t} = B^\top L\mathbf{y} \sim N(0, \sigma_g^2 \mathbf{I}_{n-p}), \qquad \mathrm{Var}\{(\mathbf{I} - QQ^\top)L\mathbf{y}\} = \sigma_g^2(\mathbf{I} - QQ^\top). \tag{21}$$

The entries of $\mathbf{t}$ are exchangeable; the n entries of the projected residual generally are not. A valid residual-coordinate transformation therefore permutes $\mathbf{t}$ and maps it back through B, rather than permuting arbitrary sample-coordinate residuals. The implementation uses orthogonal transformations without storing an additional full complement matrix in the production path. Estimated covariance, non-Gaussian errors, and repeated model selection require separate investigation; algebraic exchangeability under a known model should not be advertised as an unconditional guarantee.

A.3 BLUP scale and a marginal Bayes factor

The genetic conditional mean uses $\mathrm{Cov}(\mathbf{u}, \mathbf{y}) = \sigma_g^2 \mathbf{K}$ and $\mathrm{Var}(\mathbf{y}) = \sigma_g^2 \mathbf{H}$, hence

$$\widehat{\mathbf{u}} = \sigma_g^2 \mathbf{K}(\sigma_g^2 \mathbf{H})^{-1}(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}) = \mathbf{K}\mathbf{H}^{-1}(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}). \tag{22}$$

This identity is conditional on fitted parameters when used in the package. In-sample fitted genetic values do not measure prospective prediction accuracy.

For a marginal effect estimate $\widehat{\alpha}$ with sampling variance v, assume $\widehat{\alpha} \mid \alpha \sim N(\alpha, v)$ and an alternative prior $\alpha \sim N(0, W)$. Against a point null at zero, the approximate Bayes factor is

$$\mathrm{BF}_{10} = \sqrt{\frac{v}{v+W}} \exp\left\{ \frac{\widehat{\alpha}^2 W}{2v(v+W)} \right\}. \tag{23}$$

The prior variance W has the units of the squared effect and must be chosen explicitly. Posterior odds additionally require prior odds; normalizing marginal factors across correlated variants does not by itself yield a multi-variant fine-mapping posterior. The corrected API therefore no longer presents an RSS-based pseudo-score as a posterior probability.

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